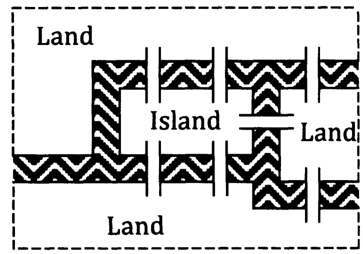
Chapter 5.

Networks.

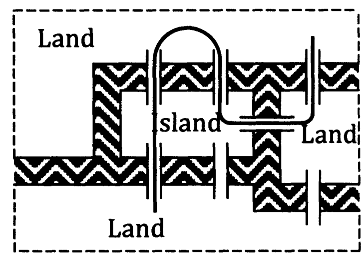
Situation One: The bridges of Konigsberg.

The city of Konigsberg, now known as Kaliningrad, features an island surrounded by the River Pregel. This island, and the other land masses involved, are connected by a number of bridges. In the 18th century the layout of the city was as shown in the diagram on the right with seven bridges connecting the various landmasses.



It is said that a popular activity of the time was to attempt to start at any of the landmasses and walk across all seven bridges, crossing each bridge once and once only.

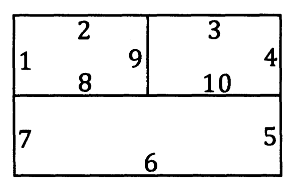
One (failed) attempt is shown on the right.



Can it be done?

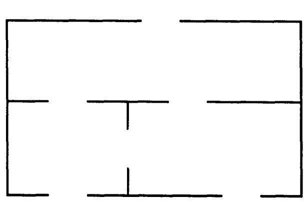
Situation Two.

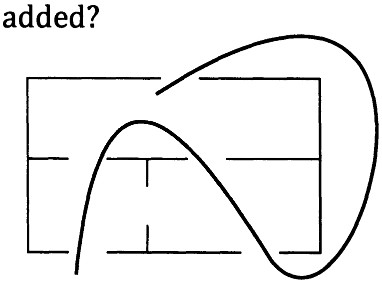
Is it possible to draw an unbroken curve to pass through all ten "walls" in the diagram on the right without passing through any wall more than once?



Situation Three.

The diagram below left shows the floor plan of a house.
The diagram below right shows one failed attempt to devise a route, starting and finishing at the same point, and that would pass through every door once and only once.
Is it possible? What if one more door could be added?





ISBN 9780170350464.

The three situations of the previous page may initially seem to have no relevance to real problems that may be faced in everyday life. However, finding the most appropriate route through a system is of great relevance in the design of such things as:

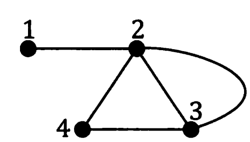
postal delivery routes,
rubbish collection routes,
house wiring systems,
computer networking systems,
routes for traffic wardens,
vehicle navigation systems, etc.

Leonhard Euler, 1707 to 1783, in explaining whether the Konigsberg bridge problem was possible or not, started the branch of Mathematics called **Topology**, of which the study of **networks** forms a part. This is not your first encounter with networks in this book because the road systems and social interactions that were mentioned in the preliminary work, and that you would have met if you studied unit one of this *Mathematics Applications* course, were presented as network diagrams.
We will return to the situations of the previous page later in the chapter but first let us consider what we mean in mathematics by a network.

Networks.
In mathematics a network is a set of points, called **vertices**, connected by a set of lines, called **edges**.

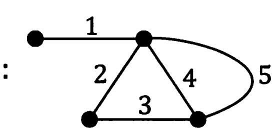
☞ The junctions and endpoints are called **vertices** (singular: **vertex**) or **nodes**.

4 Vertices:



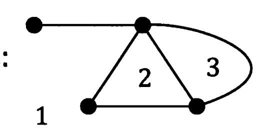
☞ The connections are called **edges**.

5 Edges:



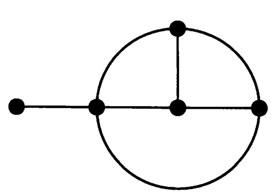
☞ The separated areas are called **faces** or **regions.** (Notice that we count the "outside" region.)

3 Regions:

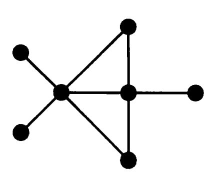


(i.e. all of "outside".)

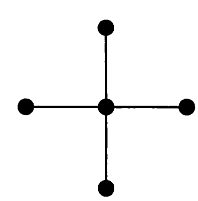
Check that you agree with each of the following:



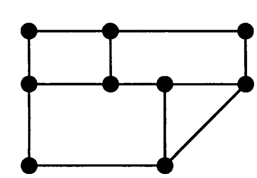
5 Vertices
7 Edges
4 Faces



7 Vertices
8 Edges
3 Faces



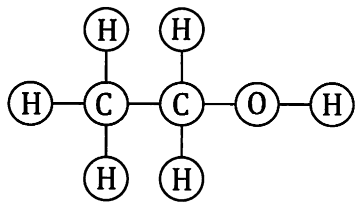
5 Vertices
4 Edges
1 Face



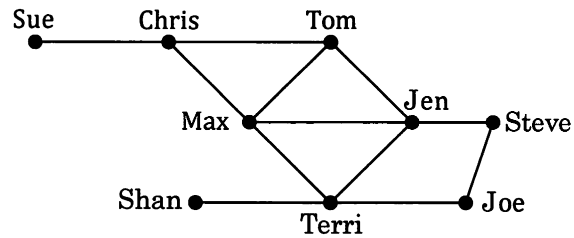
9 Vertices
12 Edges
5 Faces

Networks used in various situations are shown on the next page.

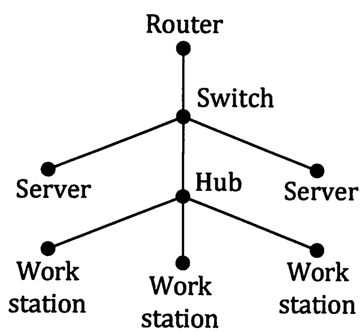
Chemical structure.



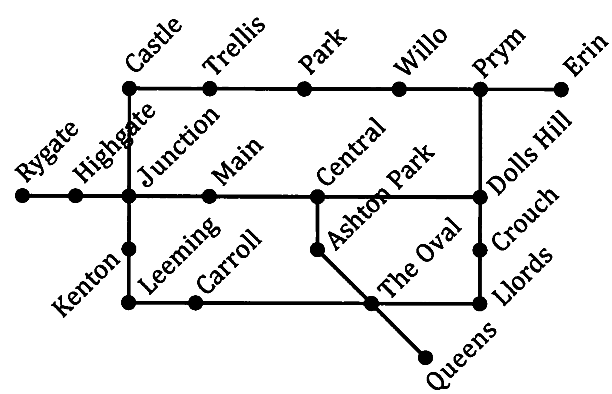
Social networks.



Technology network.



Rail system.



Exercise 5A

1. The diagram on the right is for a possible route that could be taken between Norseman and Perth.

   Various locations on the route are shown with the distances between each of these as shown.

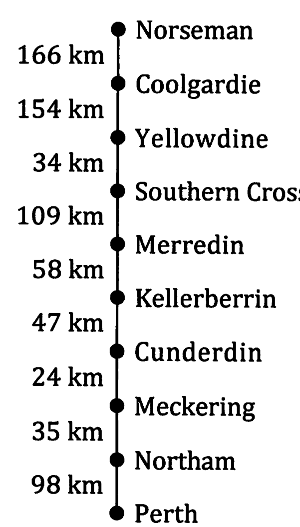
   The diagram is not drawn to scale and the route is not necessarily straight.

   For this route:

   (a) How far is it from Kellerberrin to Merredin?

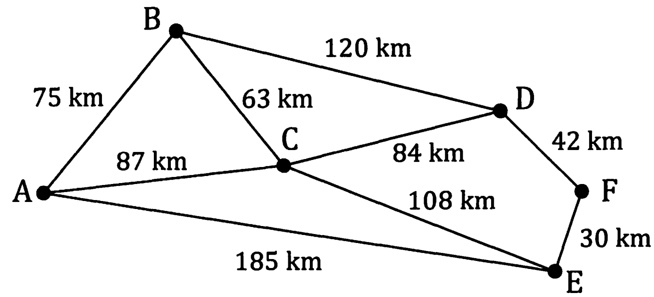
   (b) How far is it from Northam to Coolgardie?

   (c) If a steady speed of 100 km/hour could be maintained for the entire journey how long would the journey from Norseman to Perth take?



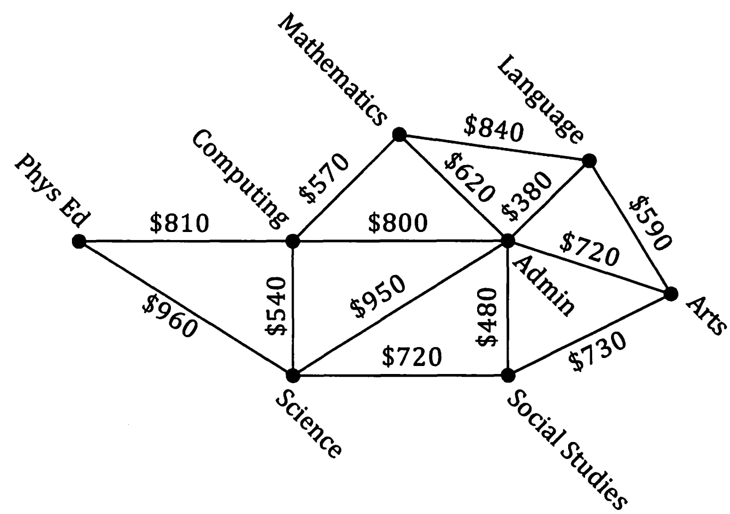
2. The diagram on the right shows the main connecting roads between six towns, A, B, C, D, E and F.

   The diagram is not to scale but distances between each town, along the given roads, are shown in kilometres.

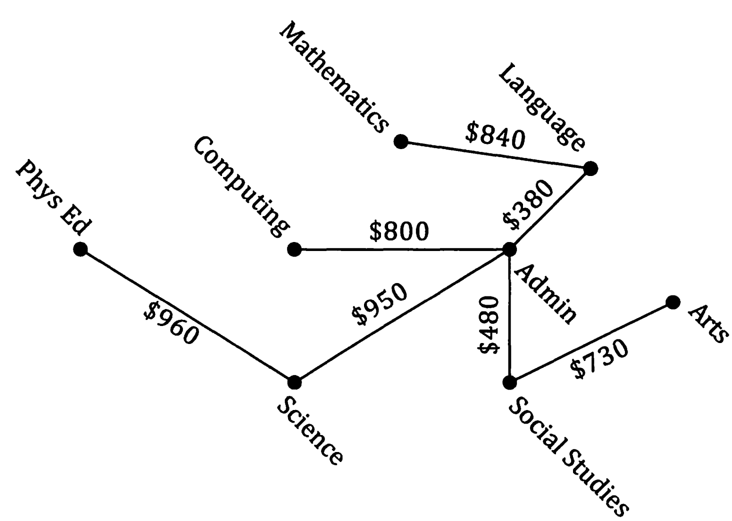


   If we must always make progress towards the right, one possible route from A to F is ABCEF, for a total distance of 276 km. List all the other routes from A to F, and state the total distance for each route, remembering all travel must be towards the right.

3. The diagram on the right shows the main buildings on a college campus, the possible cabled connections between them and the cost of each section of cabling.



   The second diagram shows one proposal for connecting each building to a central computer network by cabling.
   This proposal has a total cost of $5140.
   Try to come up with a cheaper system that still ensures that each building is "on line".
   Can you find the cheapest system that still ensures that each building is "on line"?



How many vertices, edges and faces do each of the following networks have?

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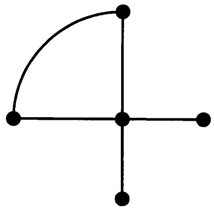

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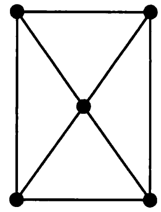

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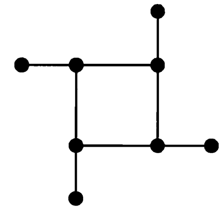

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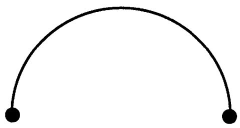

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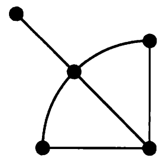

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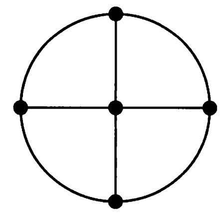

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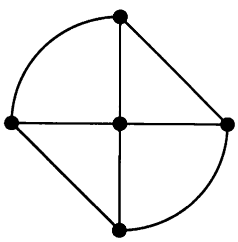

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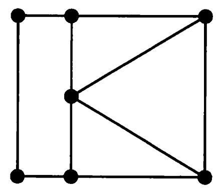

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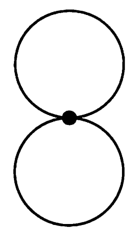

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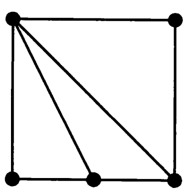

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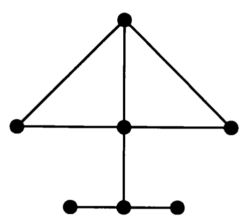

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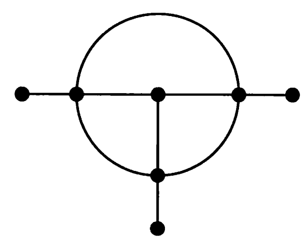

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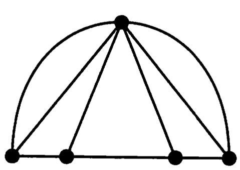

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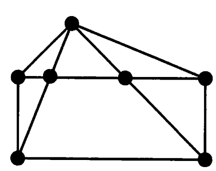

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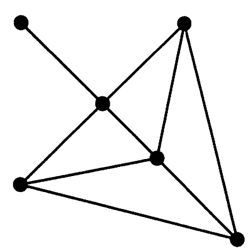

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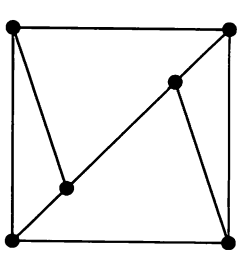

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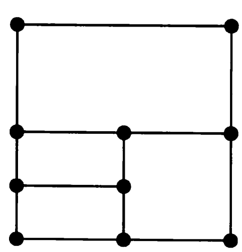

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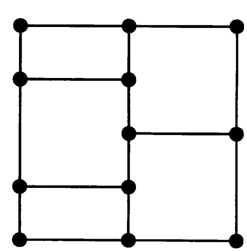


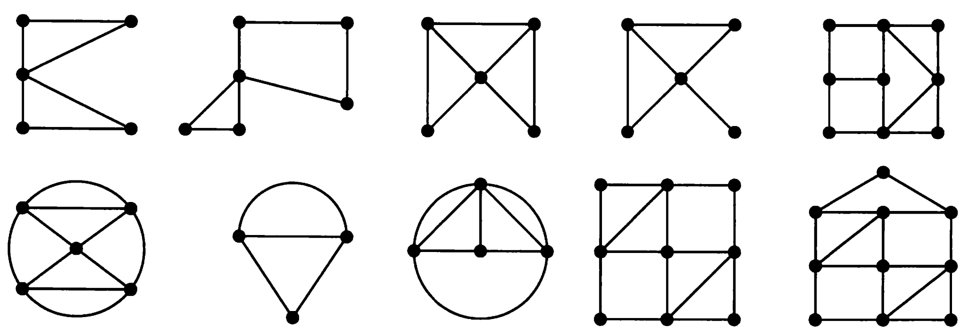
Euler's Rule.

There is a rule, called Euler's rule, connecting the number of vertices, edges and faces networks like those on the previous page have. Use your answers to questions 4 to 21 of the previous exercise to try to determine Euler's rule.

Traversability.

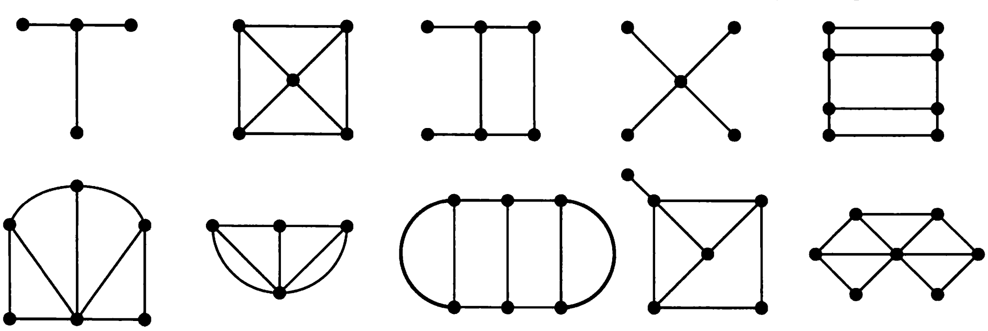
We say that a network is **traversable** if it can be drawn without taking the pen off the paper and without going over the same edge twice (you may pass through the same vertex more than once).

The following networks <u>are</u> traversable. Check that you agree - this may not be as simple as it sounds because for some it is important where you decide to start.



Some traversable networks can be traversed by starting and finishing at the same point.
Which of the traversable networks shown above have this property?

The following networks are <u>not</u> traversable. Again check that you agree.

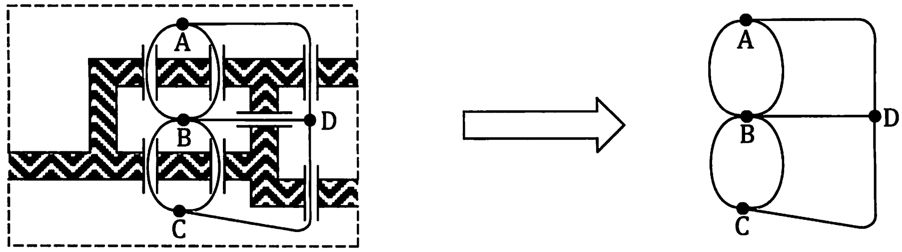


Can you discover a way of determining whether a network is traversable or not, without attempting to draw it?

Hint: Consider the number of edges meeting at each vertex.

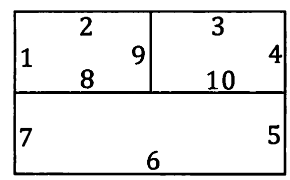
The previous investigation of traversability may have reminded you of the activities at the beginning of this chapter. Let us consider those situations again.

The first of those situations involved the bridges of Konigsberg.
If we let the land masses that must be visited to be the vertices, shown as A, B, C and D in the diagram below left, and the bridges connecting them the edges, the situation can be represented as a network as shown below right.

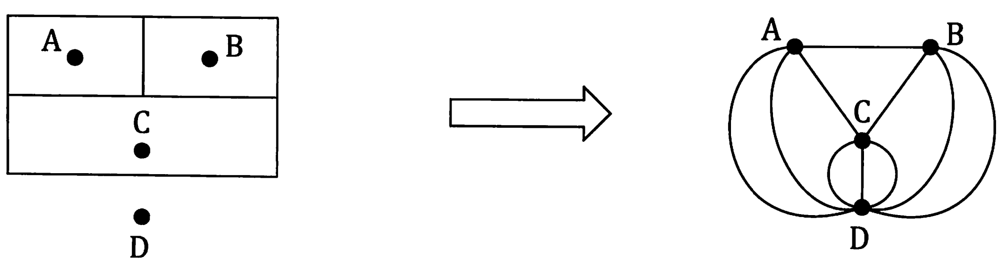


Is such a network traversable?

The second situation involved attempting to draw a curved line through all ten "walls" in the diagram on the right.

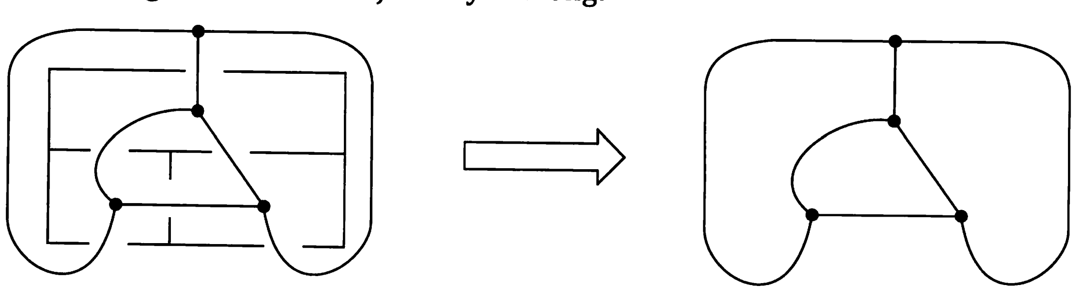


We let the separate rooms, and the outside, be the vertices, as shown below left. The routes through a wall to another vertex can then be the edges. For example if we are at A there are just two walls that if we pass through allow us to reach D without having to pass through other walls. Hence the diagram below right has two edges from A direct to D. Attempting to draw an unbroken curve through all ten walls, without passing through any wall more than once, then becomes the same as attempting to traverse the network.



Is such a network traversable?

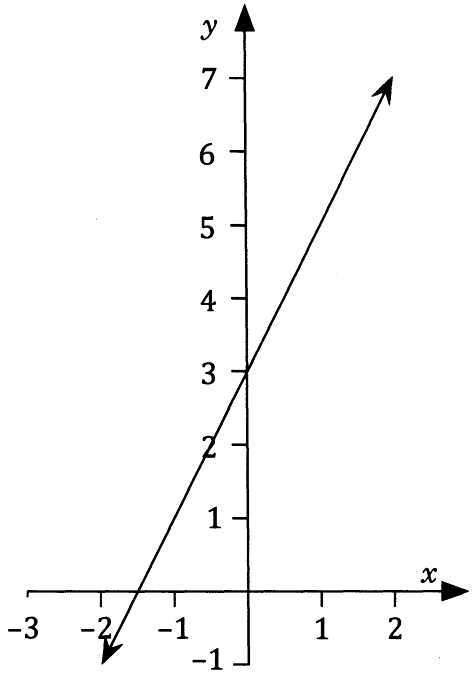
In the third situation the challenge was to find a route that would pass through every door once and only once. Let the "rooms" and "outside" be the vertices and the doorways the edges that must be journeyed along:

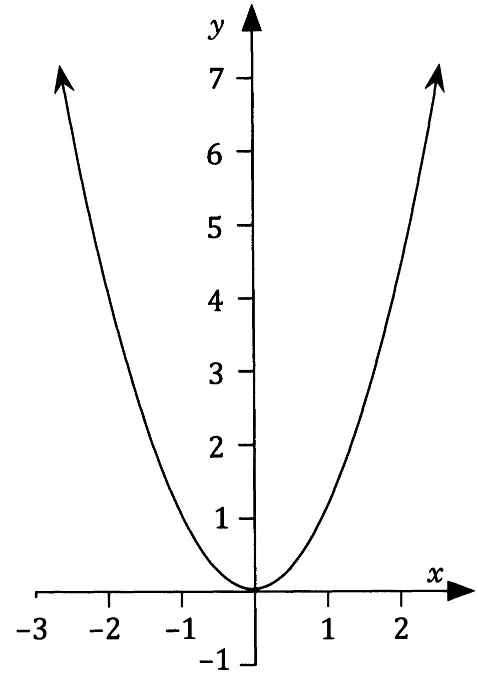


Is such a network traversable?

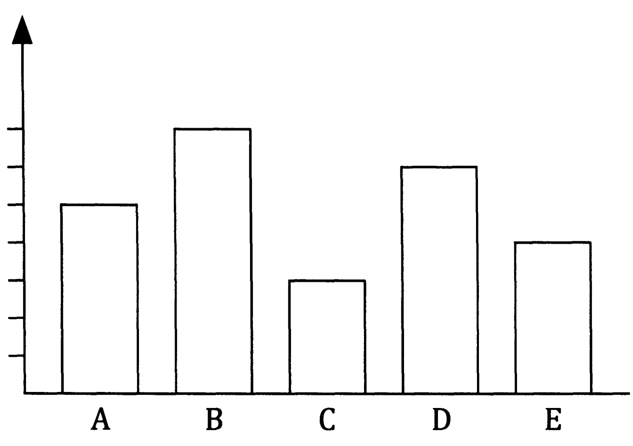
Graph theory.

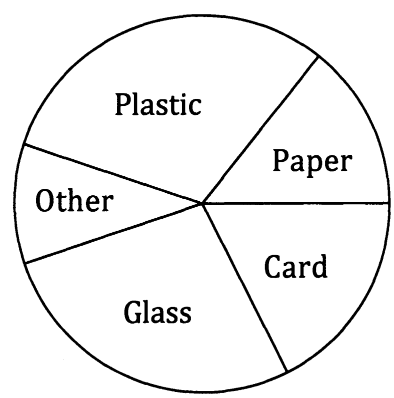
It is likely that for many students following this course a mention of the word *graph* makes you think of x and y axes, straight lines and curves, as shown below:





Perhaps some may also think of various statistical graphs:

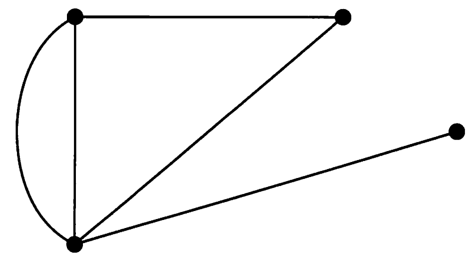




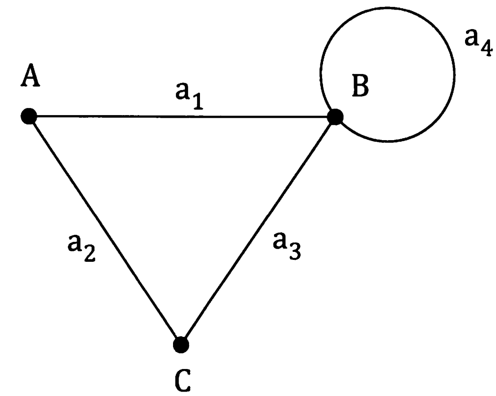
However, in this course we use the word graph in an even wider context to include structures used to display relationships and pairings. We therefore include network diagrams as graphs. Indeed the term **graph theory** tends to be taken to mean the study of networks.

Some vocabulary.

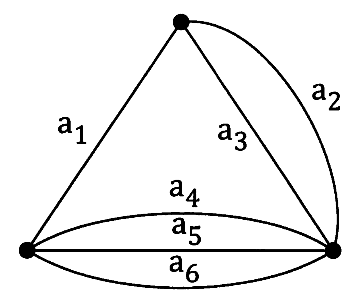
A graph is a diagram consisting of a set of points, called **vertices**, with some, or all, (or perhaps even none) of these vertices joined by a set of lines, called **edges**.



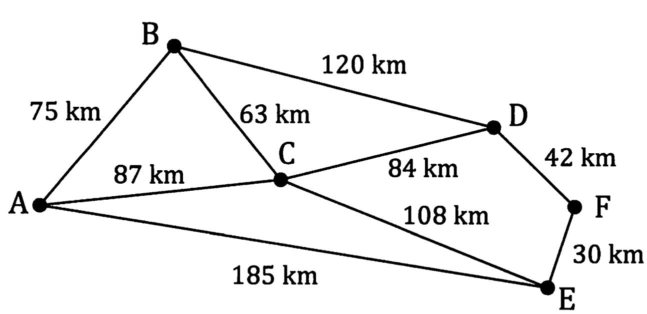
Any graph (or network) with an edge that starts and ends at the same vertex is said to contain a **loop**. In the diagram on the right edge a_4 is a loop.



If two or more edges connect the same two vertices then the edges are said to be **multiple edges**.
Thus in the diagram on the right a_2 and a_3 are multiple edges, as are a_4, a_5 and a_6.



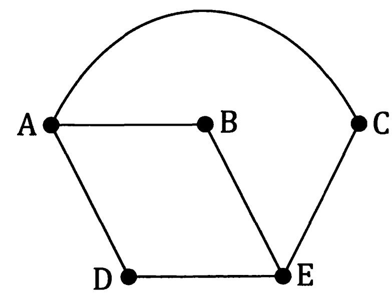
Some of the networks encountered earlier in this chapter had amounts or distances or some information on each edge, for example:



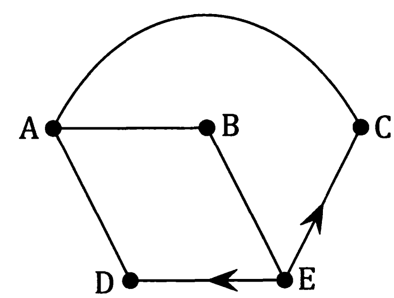
This is an example of a **weighted graph, or weighted network**. Each edge is worth a particular *weight*, be it distance, cost, time etc.

As the preliminary work section at the beginning of this text mentioned, if you have previously completed *Unit One* of *Mathematics Applications* you will be familiar with the idea of road systems and social interactions being displayed in the form of a network.

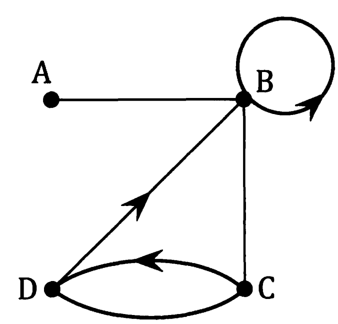
Suppose that the social interaction network, or graph, on the right shows which people, in a group of five people, A, B, C, D and E, have *shaken hands with* each other.
The network indicates that A has shaken hands with B, C and D but not E. C has shaken hands with A and E but not with B or D.
The action of shaking hands is directionless. If A has shaken hands with C then C has shaken hands with A.
Hence there are no directions shown on the network.



Suppose instead that the interaction network shown on the right shows who, in the group of five, has *been to the house of* someone else in the group. Now direction has significance. As the graph shows, E has been to the house of C but C has not been to the house of E. It makes sense for the graph to involve some **directed edges,** also referred to as **arcs.**



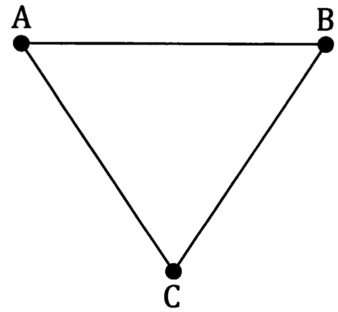
Similarly in the network of roads that exist linking towns A, B, C and D, as shown on the right, the arrows indicate that some roads are one way only.
Again it makes sense for the graph to involve some directed edges.



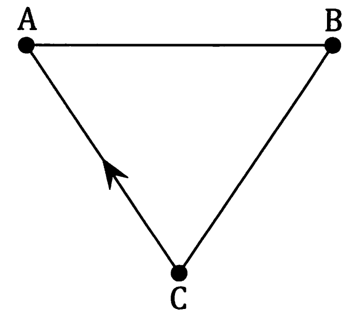
Any graph having no directed edges is an **undirected**, or **non-directed**, graph.

Any graph involving directed edges is called a **directed graph,** also called a **digraph.**

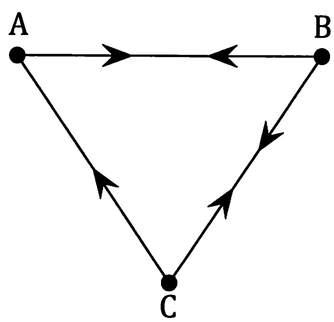
An undirected graph.

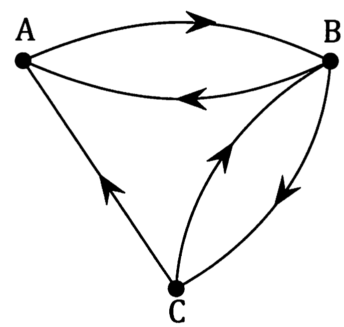


A directed graph.



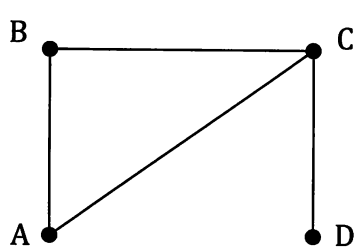
Note • In this unit, now that we are making the distinction between undirected graphs and directed graphs (digraphs), we will tend to show directions on all of the edges of a directed graph, rather than leave some "unarrowed" as in the diagram above right. Hence in this book the digraph shown above right will tend to be drawn in one of the forms shown below.



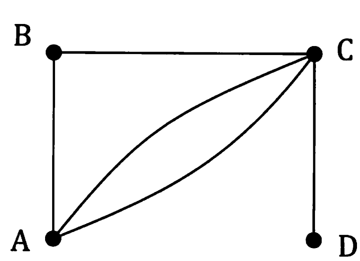


An undirected graph could be shown as "directed" by placing opposing arrows on every edge, for example to show that all roads in a system are "two way". However such situations are more simply shown as undirected graphs.

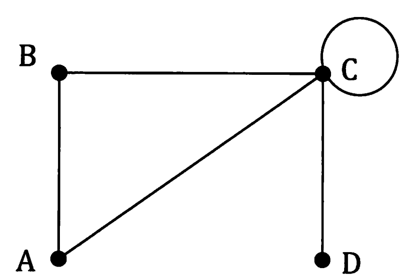
If we refer to a **simple graph** or **simple network** we mean an undirected and unweighted graph with no loops and no multiple edges.



A simple graph



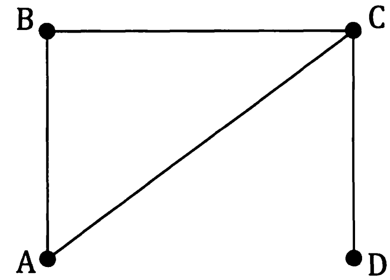
Not a simple graph.
(Has multiple edges.)



Not a simple graph.
(Has a loop.)

Note: Directed graphs and weighted graphs having no loops or multiple edges are sometimes referred to as **simple directed graphs** and **simple weighted graphs**. (Multiple edges in a digraph have the same start point and the same end point.)

Vocabulary continued - let's take a walk.



Notice that in the graph, or network, shown on the right we could "take a walk" from A to D via C. In graph theory, such a sequence of vertices, in this case A - C - D, in which there is an edge from each vertex to the next, is indeed called a **walk**.
In this network, ACD is not the only walk there is between A and D. Another would be ABCD.

Any walk that starts and finishes at the same vertex is called a **closed walk** whilst not ending at the starting vertex indicates an **open walk**.

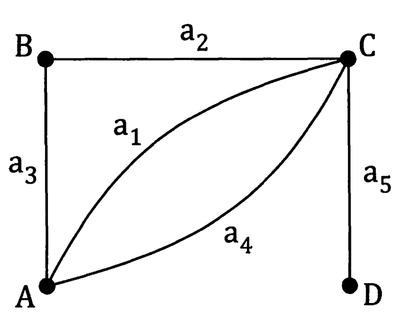
For the network above right, ABCD is an open walk whilst ABCA is a closed walk.

A walk may include repeated vertices and repeat use of edges, eg the walk ABCDCBC in the graph above, but if there is no repeat use of edges *and* no repeat use of vertices, other than perhaps ending at the same vertex as we start from, the walk is referred to as a **path.**

Thus for the network shown, ABCD is a **path** between A and D. In fact, to be more informative, ABCD is an **open path** as the starting vertex and the finishing vertex are different. In contrast, ABCA is an example of a **closed path** or **cycle**, because now the path finishes back at the starting vertex.

The "**length**" of a walk, path or cycle is the number of edges the walk, path or cycle uses. Thus the walk A - B - C - D - C - B - C has *length* 6 (repeat use of an edge counting each time), the path A - C - D has *length* 2 and the cycle A - B - C - A has *length* 3.

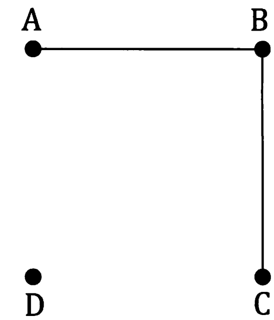
If a walk involves no repeated edges then we have a **trail**. In the network above, of the walks already mentioned, ACD, ABCD and ABCA are all trails but ABCDCBC is not due to the repeat use of some edges. However, while all paths are trails (because to be a path means no repeat use of edges and no repeat use of vertices whereas to be a trail only requires no repeat use of edges) we can have a trail that is not a path. For example, in the graph below, if we follow the edges in the sequence $a_1\ a_2\ a_3\ a_4\ a_5$ then ACBACD is a trail, but the repeat use of vertices A and C means that it is not a path.



If a trail ends at the vertex it started from it is a **closed trail**.

Even more vocabulary – let's check the connections.

In an undirected graph, if a path exists from one vertex, A, to another vertex, B, even if that path involves travelling along several edges, then the vertices A and B are said to be **connected vertices**. If the two vertices are connected by a path that does indeed only involve one edge then the vertices are said to be **adjacent vertices**. Hence adjacent vertices share a common edge.



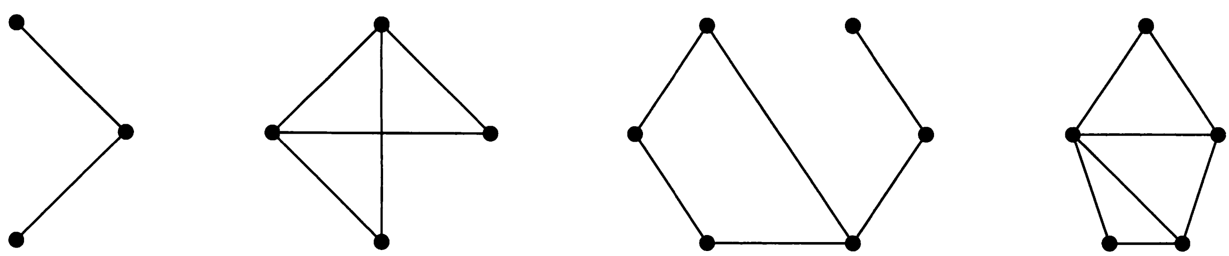
Thus in the graph shown on the right:

☞ A and B are connected vertices
A and C are connected vertices
B and C are connected vertices

☞ A and B are adjacent vertices
B and C are adjacent vertices.

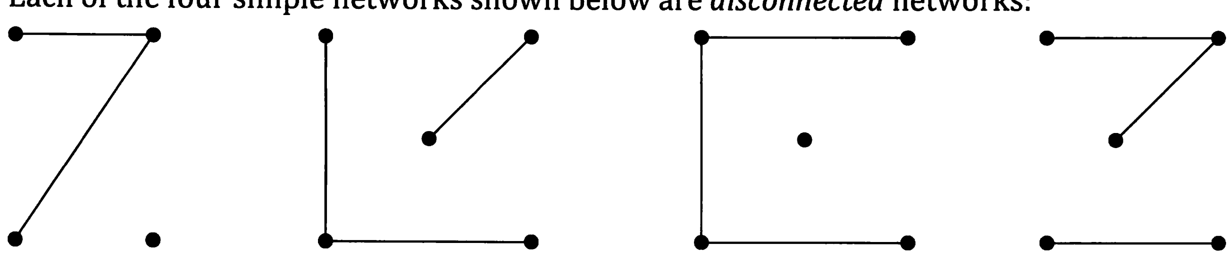
If, in an undirected graph (or network), every vertex is connected to every other vertex then the graph (or network) is said to be a **connected graph** (or **connected network**). Hence in the network above right, with there being no path from A to D (or from B to D or from C to D) the network is not a connected network. It is said to be **disconnected** or **unconnected**.

Each of the four simple networks shown below are *connected* networks:



Important note • The second graph of the four shown above is the first in this chapter that has shown edges crossing at a point that is not marked as a vertex. This is fine, simply imagine that one edge is passing over (or under) the other. If the edges were pipes or wires one pipe or wire passes above the other. In this way the graph is a two dimensional representation of a three dimensional situation. We will consider this aspect again later.

Each of the four simple networks shown below are *disconnected* networks:



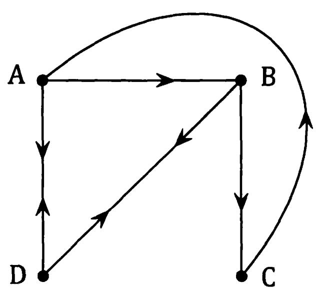
Note: With directed networks the situation is a little different and we would instead talk about the "strength" of the connection. A search of the internet reveals a number of different conventions for describing connectedness in directed graphs. One system describes a directed network as being strongly connected, connected or weakly connected according to the following criteria:

Strongly connected if for every pair of vertices there exists a path from one vertex to the other AND a path back.

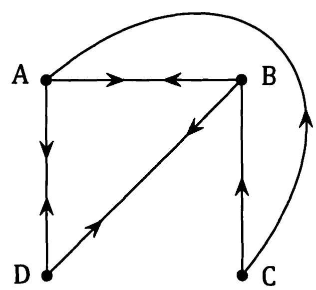
Connected if for every pair of vertices there is at least one path connecting the vertices, even if "a way back" is not possible.

Weakly connected if replacing all of the directed edges by undirected edges produces a connected undirected network.

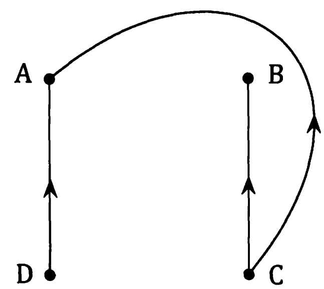
Under this system, a strongly connected directed network is also connected and weakly connected. However we would tend to quote the strongest category and say it is strongly connected. Similarly a connected directed network is also weakly connected but we would simply say it is connected.



A strongly connected directed network.

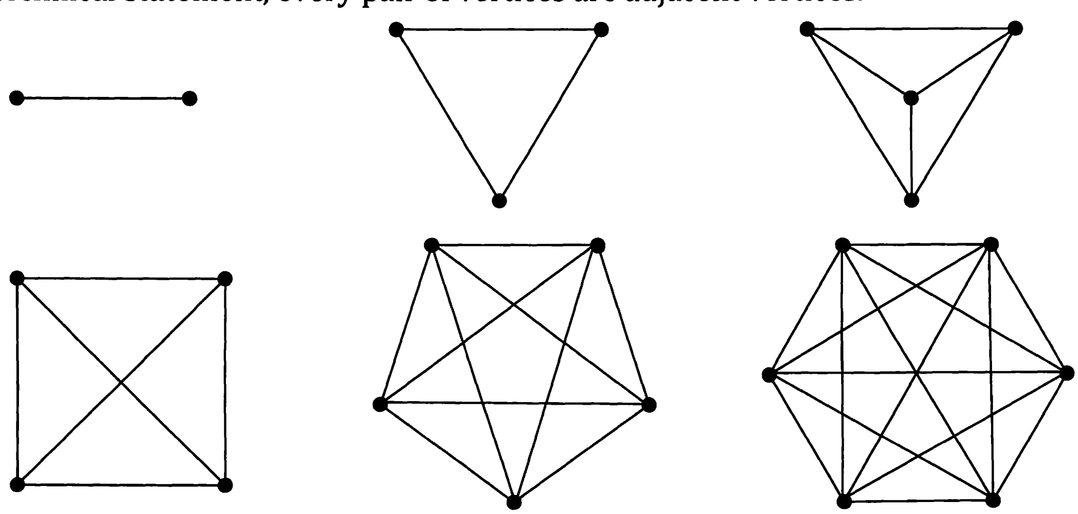


A connected directed network.



A weakly connected directed network.

Notice that in the six simple connected networks shown below each vertex is connected to *every* other vertex in the network by a single edge. Or, to put it another way, we can make a journey from each vertex, along a single edge, to every other vertex. Or, to give a more technical statement, every pair of vertices are adjacent vertices.

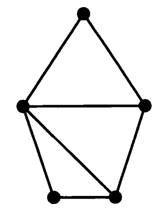


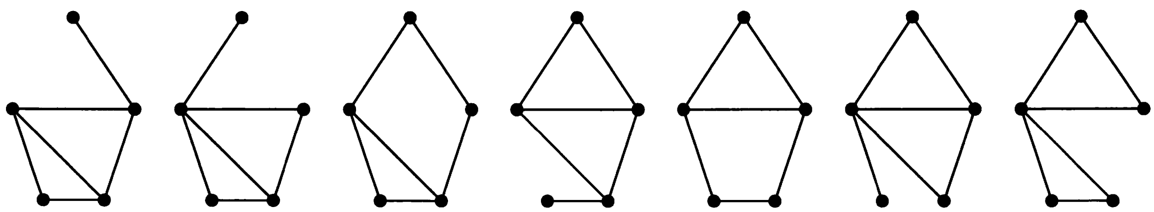
➔ Simple graphs in which every vertex is connected to every other vertex by, in each case, a single edge, are called **complete** graphs. 🡐

Note: A complete graph with n vertices is sometimes denoted as K_n.
The previous page shows K_2 to K_6. (K_4 being shown twice.)

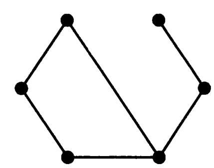
Consider the connected graph shown on the right.

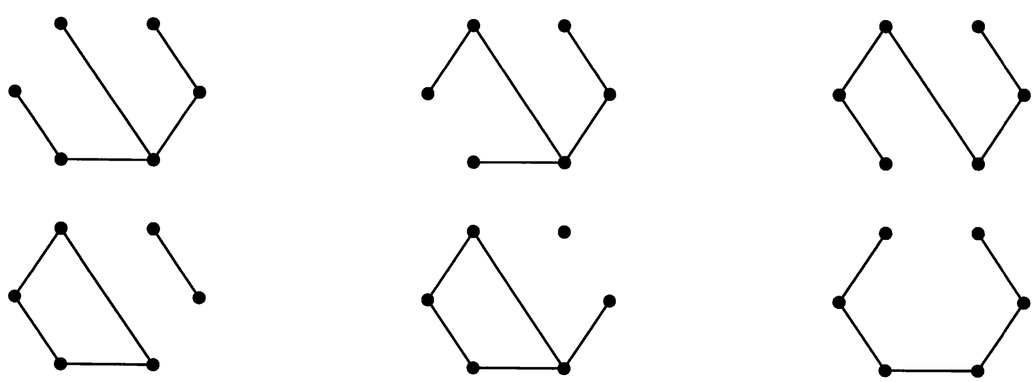
Notice that we can remove any one of the seven edges in this graph and in each case the graph remains connected:





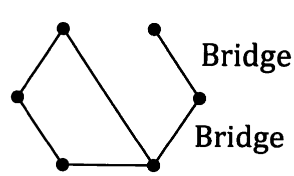
Contrast this with the effect of the same single edge removal carried out on the connected graph that is now on the right.
(Note: The removal of an edge leaves all vertices in place.)





Notice that now, in some cases, removal of a single edge leaves the graph disconnected.

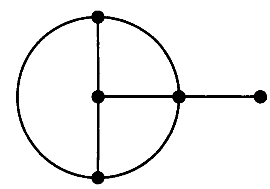
In a connected graph any edge which, when removed, leaves the graph disconnected, is called a **bridge**.



Summary!

Wow, the previous pages introduced a lot of new vocabulary! Whilst you should have managed to understand each word or expression as it was introduced you may now be having difficulty remembering exactly what each meant. Read through the following summary to revise the new vocabulary.

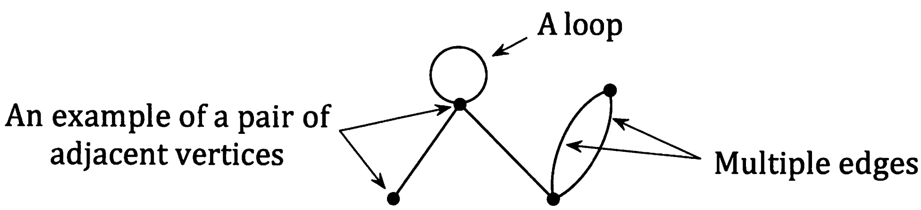
Graph theory involves the study of diagrams, called **networks** or **graphs**, consisting of a set of points, called **vertices** with some, or all, (or perhaps even none) of these vertices joined by a set of lines, called **edges**. These edges can divide the graph up into a number of separate **faces** (or **regions**).



Any graph with an edge that starts and ends at the same vertex is said to contain a **loop**.

If two or more edges connect the same two vertices then the edges are said to be **multiple edges**. (If the two edges have direction then they are only multiple edges if they have the same direction between the same two vertices.)

If two vertices are connected by an edge then the vertices are said to be **adjacent vertices.**



Graphs that have amounts or distances or some information on each edge are referred to as **weighted graphs**, or **weighted networks**.

Graphs that have **directed edges**, or **arcs**, are called **directed graphs** or **digraphs**.

Graphs with no directed edges are referred to as **undirected graphs**.

An undirected, unweighted graph with no loops and no multiple edges is called a **simple graph**. (Digraphs and weighted graphs with no loops and no multiple edges are sometimes referred to as **simple digraphs** and **simple weighted graphs**.)

A **walk** in a graph is a sequence of vertices for which each vertex in the sequence is joined to the next vertex in the sequence by an edge.

Any walk that starts and finishes at the same vertex is called a **closed walk** whilst not ending at the starting vertex indicates an **open walk**.

A walk may include repeated vertices and repeated edges.

If a walk involves no repeat use of edges and no repeat use of vertices, other than perhaps ending at the same vertex as we start from, the walk is referred to as a **path.**

A path that starts and finishes at different vertices is said to be an **open path**.

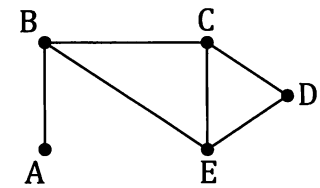
A path that starts and ends at the same vertex is said to be a **closed path** or **cycle**.

The "**length**" of a walk is the number of edges it uses (repeat use of edges counted each time).

If a walk involves no repeated edges then we have a **trail**. However, while all paths are trails (because to be a path means no repeat use of edges and no repeat use of vertices whereas to be a trail only requires no repeat use of edges) we can have a trail that is not a path (if for example the trail involves a repeat vertex).

If a trail ends at the vertex it started with it is a **closed trail**.

Hence, for the graph shown on the right, the sequence of vertices AEC is not a walk (and therefore neither path nor trail) because there is no edge from A to E.



For this graph, whilst the following sequences of vertices are all walks ABCD, BCDEB, CDECBA, it would be more informative to say that

ABCD is an open path, of length 3.
(No repeated vertices, no repeated edges, hence a path.)

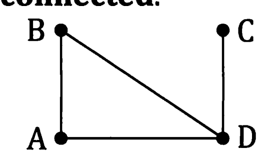
BCDEB is a closed path, or cycle, of length 4.
(No repeat vertices, apart from same start and finish, and no repeat edges.)

CDECBA is an open trail, of length 5.
(No repeat edges but a repeated vertex, hence a trail not a path.)

In an undirected graph, if a path exists from one vertex, A, to another vertex, B, even if that path involves travelling along several edges, then the vertices A and B are said to be **connected vertices**.

If, in an undirected graph, every vertex is connected to every other vertex then the graph is said to be a **connected graph**. If a graph is not connected it is **disconnected**.

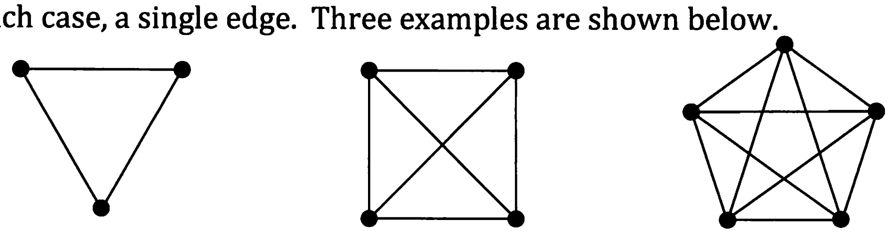
In the graph shown on the right, paths exist from each vertex to every one of the other vertices. Hence every vertex is connected to every other vertex and so the graph is connected.



Any edge in a connected graph which, when removed, leaves a graph disconnected is called **a bridge**.

[With directed networks, or digraphs, the situation with regards to how the graph is connected is a little different. Instead we talk about the "strength" of the connection as explained on an earlier page.]

Complete graphs are simple graphs in which every vertex is connected to every other vertex by, in each case, a single edge. Three examples are shown below.

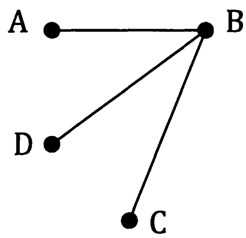


A complete graph with n vertices is sometimes denoted as K_n.

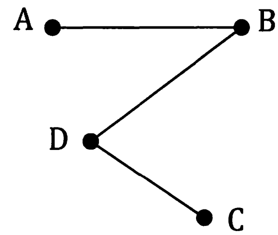
Exercise 5B

1. For each of the following state whether the graph is connected or disconnected. (For (d) and (e) remember the "important note" on page 123 regarding edges passing over, or under, each other.)

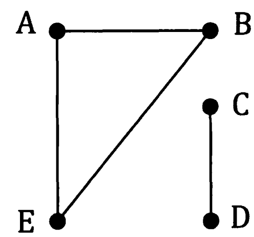
(a)



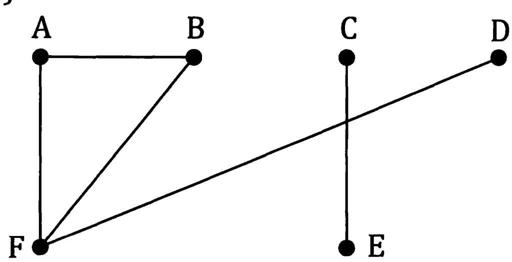
(b)



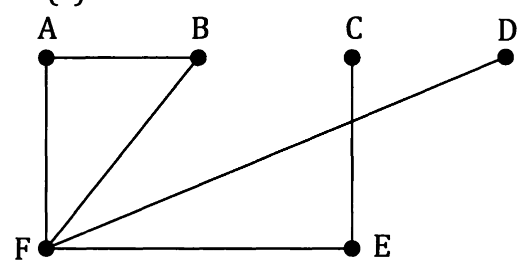
(c)



(d)

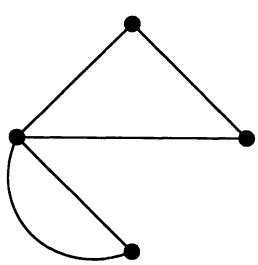


(e)

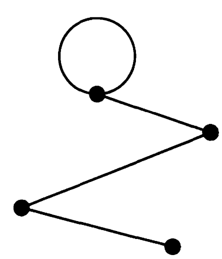


2. Defining a simple graph as an undirected, unweighted graph with no loops and no multiple edges, only two of the six graphs shown below are *simple* graphs.
State which two they are.
For each of those that are not simple graphs explain why not.

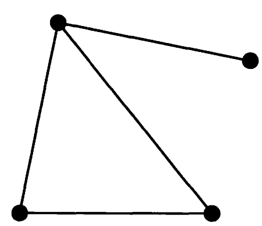
(a)



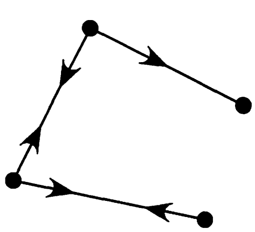
(b)



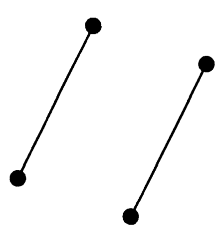
(c)



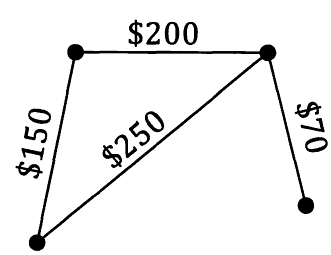
(d)



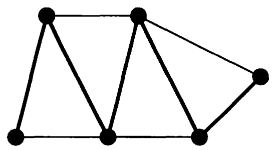
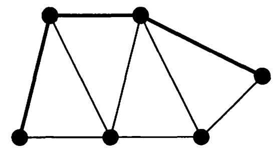
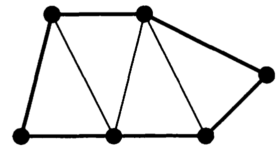
(e)



(f)



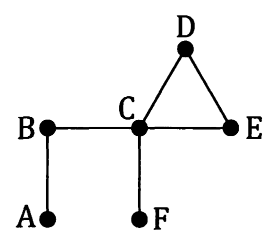
3. State the length of the paths and of the cycle shown below in heavy type.

(a)  (b)  (c) 

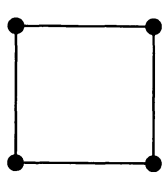
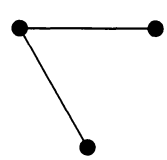
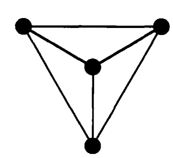
4. For each of the situations below state which is likely to be the more appropriate form of representation - an undirected graph or a directed graph (a digraph) and explain your reasoning.
   (a) A graph showing "who likes who" for a group of people.
   (b) A graph showing "who is aware of the existence of who" for a group of people.
   (c) A graph showing "who is a Facebook friend of who" for a group of people.
   (d) A graph showing "who is related to who" for a group of people.
   (e) A graph showing "who has been to the movies with who" for a group of people.
   (f) A graph showing "who lives within 5 km of who" for a group of people.
   (g) A graph showing "who has the phone number of who" for a group of people.

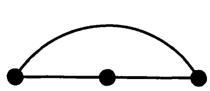
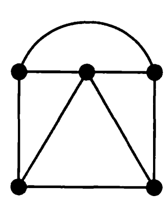
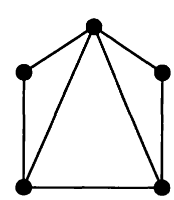
5. (a) Explain why the edge CE in the connected graph shown on the right is <u>not</u> a bridge.

   (b) State which edges in the network shown are bridges and justify your answer.



6. Which of the following simple graphs are complete graphs?

(a)  (b)  (c) 

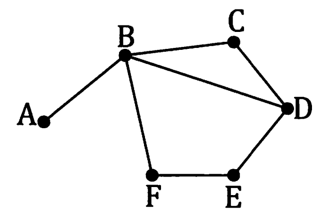
(d)  (e)  (f) 

7. We denote a complete graph involving n vertices as K_n.
Draw K_4 and K_8.

8. (a) For the network shown on the right, state which of the following sequences of vertices are walks.

AFE ABCDE BCD ABDEF

ABDFA ABDEFBC BDEFB

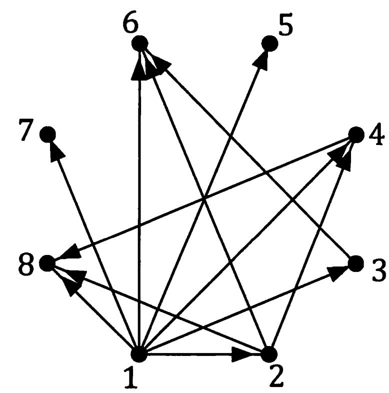


(b) For each of the above sequences of vertices that are walks, describe each as one of

a path, or a cycle, or a trail,

whichever is the more informative in each case.

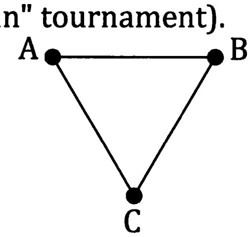
9. The graph on the right shows the relationship
is a proper divisor of
for the integers 1 to 8.
Construct similar graphs showing
is a proper divisor of
for (a) the integers 1 to 12,
(b) the integers 2 to 16.



10. Consider a tennis tournament involving three players in which each player plays a singles match against each of the other players (i.e. a "round robin" tournament). If we denote the players as A, B and C the tournament will involve the following three matches:

A vs B, A vs C, B vs C.

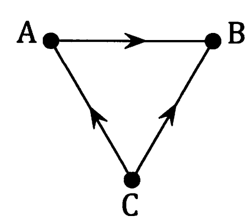
This three match system can be represented by the complete graph K_3 shown on the right.



Let us suppose that the results of these matches were:

A beat B C beat A C beat B

Adding directions to the previous graph allows these results to be included on the graph, as shown on the right.



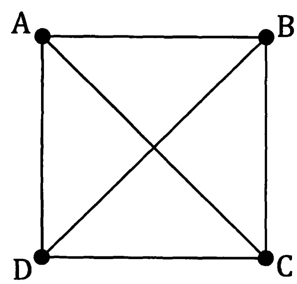
Use this idea to show the following results from a five player round robin as a digraph, with the direction of the arrow indicating who beat who.

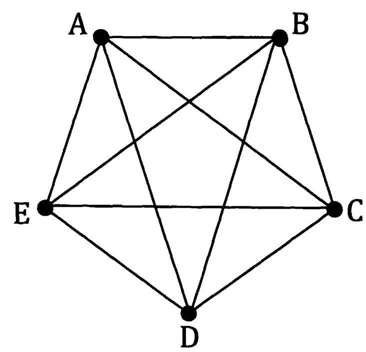
A beat B	C beat A	A lost to D	E lost to A	C beat B
B lost to D	B beat E	C beat D	C lost to E	D beat E

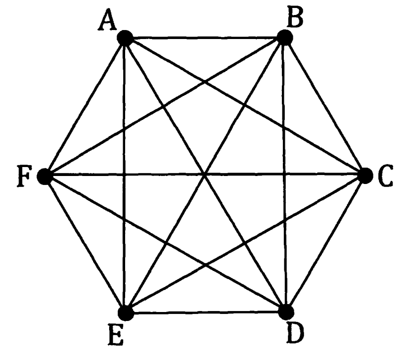
Planar graphs, Euler's rule, subgraphs, trees and bipartite graphs.

Planar graphs.

Let us consider again the three complete graphs shown below that featured on an earlier page.



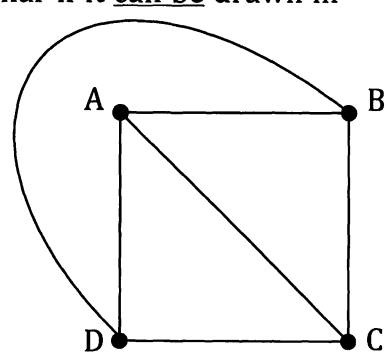




Notice that each of these graphs feature edges "crossing over each other". Where the edges cross are not vertices but instead they are places where one edge "passes over" another. If these networks were, for example, systems of water pipes or wires in an electrical circuit, these locations would be where one pipe or wire passes over another. This "one pipe above the other" situation means that the network has a place or places where the edges are not all "in the same plane".

We say that a graph that *can be* drawn in the plane, without its edges crossing over, is **planar**.

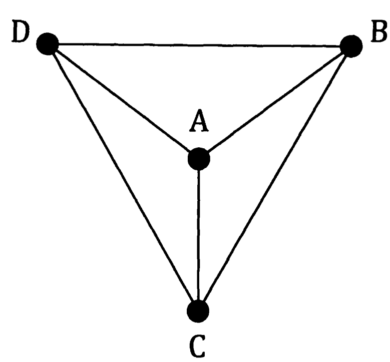
An important point to note here is that we say a network is planar if it can be drawn in the plane with its edges only intersecting at vertices.

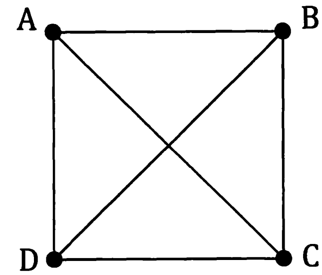


The square shaped network in the three shown above is planar because, whilst it hasn't been drawn without edges crossing over, it can be, as shown on the right.

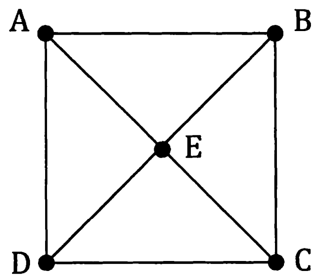
Drawn in this way the network is more obviously planar.

As we saw earlier, this same planar network could be drawn as shown below.





Consider again the network shown on the right.
In some situations it may be possible to insert a junction or vertex at the "cross over point", as shown below:



However, whilst the new system is more obviously *planar* it is no longer *complete* because there is now no single edge linking A and C (nor B and D). Also, in a real situation involving water pipes or electrical circuits, we may need AC and BD to be separate pipes or wires that do not form a junction with each other. Thus simply adding more vertices to give a planar system can change the original system significantly, especially in some real life contexts.

Euler's Rule.

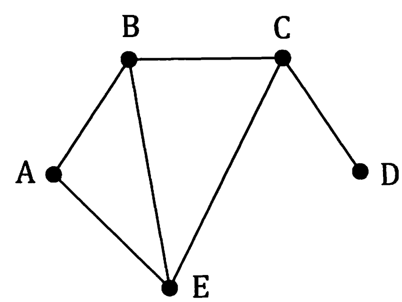
Earlier in this chapter you were asked to determine Euler's rule using your answers for the number of vertices, edges and faces a number of graphs had. Did you manage to discover the rule?
The rule, for any connected planar network, is that if v is the number of vertices, f the number of faces and e the number of edges then:

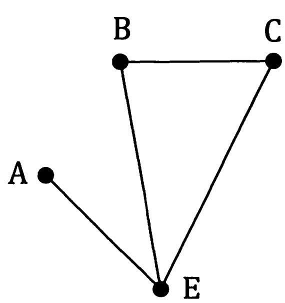
$$v + f = e + 2$$

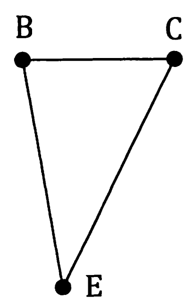
(I.e.: vertices + faces = edges + 2.)

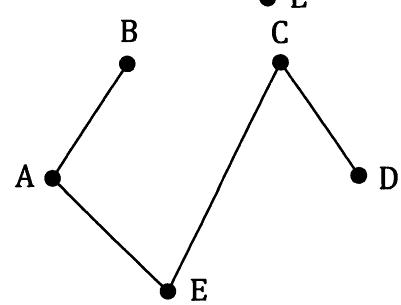
Subgraphs.

Notice that each of the three graphs shown below involve a selection of the vertices and edges from the graph shown on the right.





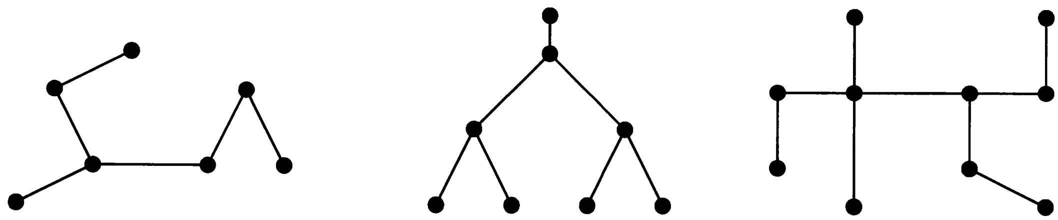




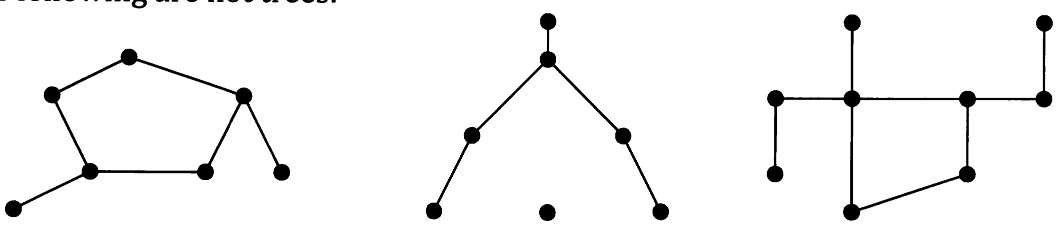
These three are all **subgraphs** of the original graph. (And indeed the middle graph of the three subgraphs is itself a subgraph of the one on the left.)

Trees.

Any connected simple graph that contains no cycles is called a **tree**.
The following graphs are trees:



The following are not trees:

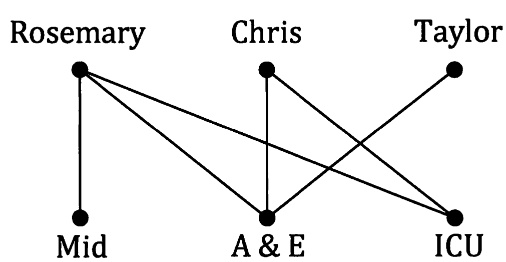


Contains a cycle. Therefore not a tree.	Not connected. Therefore not a tree.	Contains a cycle. Therefore not a tree.

Note: A tree with n vertices will have (n – 1) edges.
In a tree every edge is a bridge.

Bipartite graphs.

The graph below shows which of the three areas, midwifery (Mid), accident and emergency (A & E), and the intensive care unit (ICU), that three nurses are qualified to work in.



If we consider the vertices of this graph to be in two groups, Rosemary, Chris and Taylor being one group and Mid, A&E and ICU the other, each edge joins a vertex from one group to a vertex from the other group. There is no edge that by itself joins two vertices from the same group.
A graph in which the vertices can be split into two groups such that every edge joins a vertex from one group to a vertex of the other group, i.e. no edges join vertices from the same group, is called a **bipartite** graph. Thus, in a bipartite graph, every pair of adjacent vertices involves one vertex from one group and the other vertex from the other group.
(If a bipartite graph has every member of each group connected to every member of the other group it is a *complete bipartite graph*.)

Exercise 5C

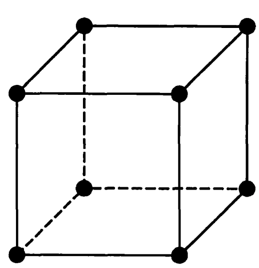
1. A connected planar graph has 5 vertices and 3 faces. How many edges does the graph have?

2. A connected planar graph has 4 faces and 6 edges. How many vertices does the graph have?

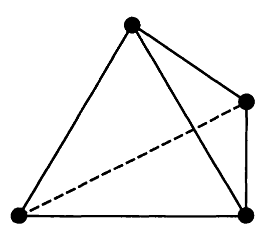
3. A connected planar graph has 5 vertices and 12 edges. How many faces does the graph have?

4. In this work on graph theory we are using the words vertices, edges and faces. However you have probably long been familiar with the use of these words when describing the features of some three dimensional shapes.
With the words used in this geometric, rather than graphical, sense determine the number of vertices, edges and faces each of the following three dimensional shapes have and then see if Euler's rule (Vertices + Faces = Edges + 2) is obeyed for each one.

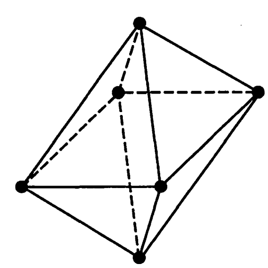
(a) A cube



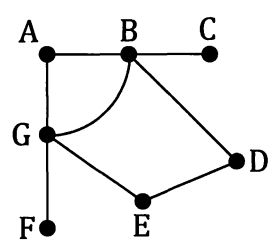
(b) A regular tetrahedron.
(A polyhedron with four faces, each of which is an equilateral triangle.)



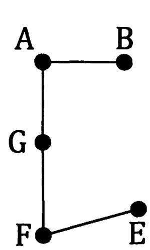
(c) A regular octahedron.
(A polyhedron with eight faces, each of which is an equilateral triangle.)



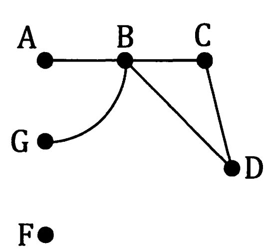
5. Which of the graphs shown below are subgraphs of the graph shown on the right?
For any that are not explain why.



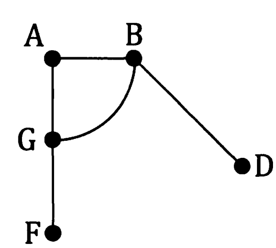
(a)



(b)



(c)

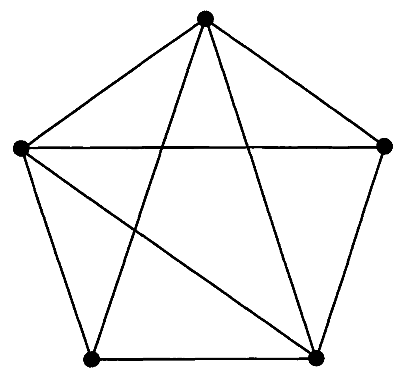


6. The table below shows which soccer clubs out of Manchester United, Manchester City, Liverpool, Everton and Arsenal, four soccer players, PB, BK, MO and FS played for (with a tick indicating the player did play for that club and a cross indicating the player did not).

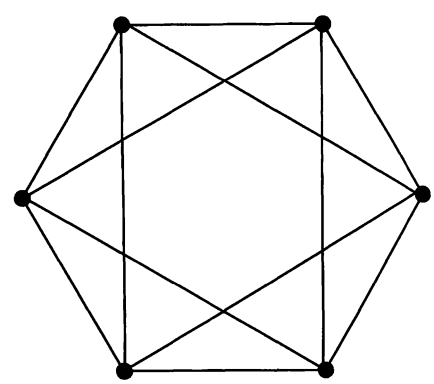
	Arsenal	Everton	Liverpool	Manchester City	Manchester United
PB	✗	✔	✔	✔	✔
BK	✔	✔	✗	✔	✔
MO	✗	✗	✔	✗	✔
FS	✔	✗	✗	✗	✔

Show the information in the form of a bipartite graph.

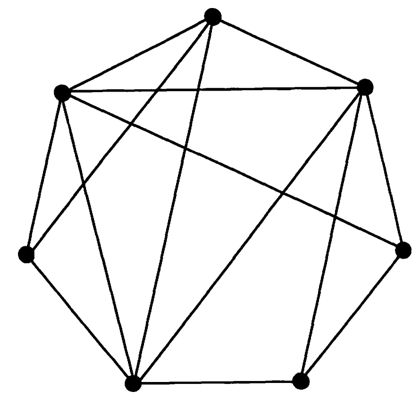
7. For the graph shown on the right:
   (a) Explain why the graph is not a complete graph.
   (b) Show that the graph is planar by producing a version of it in which the edges only meet at vertices.



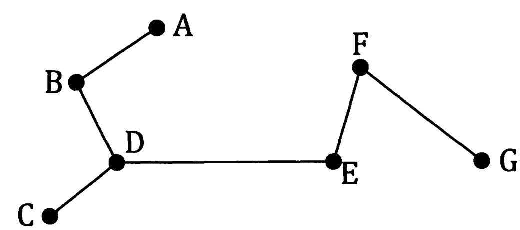
8. For the graph shown on the right:
   (a) State whether the graph is a complete graph or not and explain your reasoning.
   (b) Show that the graph is planar by producing a version of it in which the edges only meet at vertices.



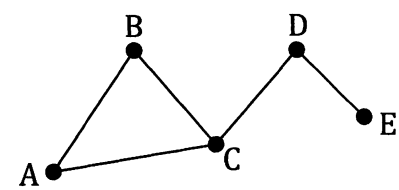
9. Show that the network drawn on the right is planar by producing a version of it in which the edges only meet at vertices.



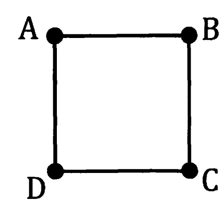
10. Given that the tree shown on the right is a bipartite graph what are the two vertex sets? Draw the graph as a more obviously bipartite graph.



11. Is the graph shown on the right a bipartite graph? Justify your answer.

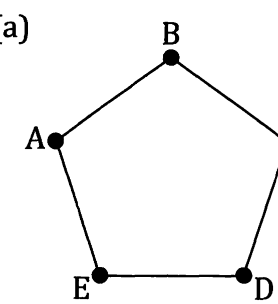


12. Is the graph shown on the right a bipartite graph? Justify your answer.

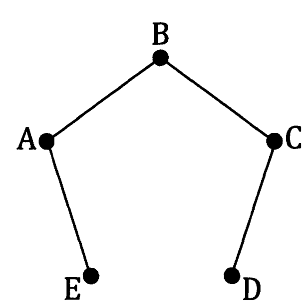


13. For each of the graphs shown below state whether the graph is bipartite.
If you claim that it is not bipartite justify your claim.
If you claim that it is bipartite redraw it as a more obviously bipartite graph.

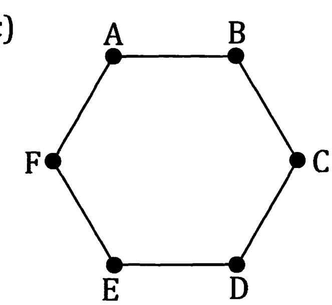
(a)



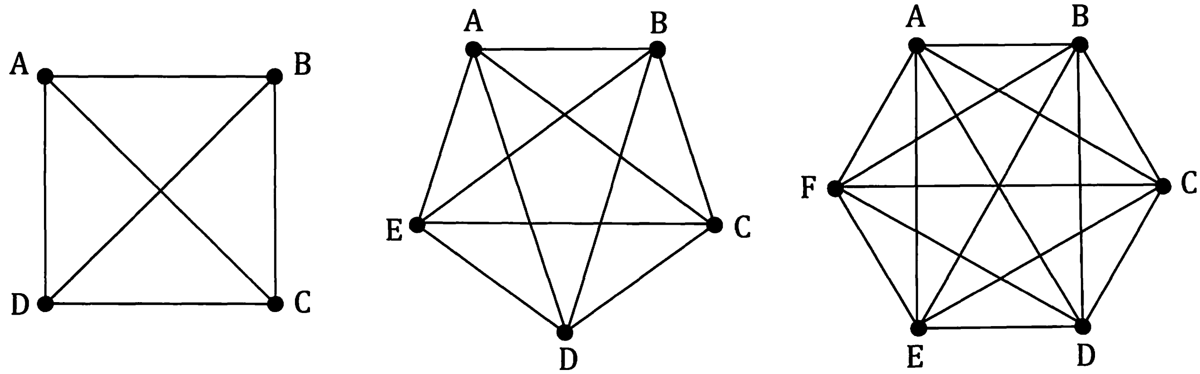
(b)



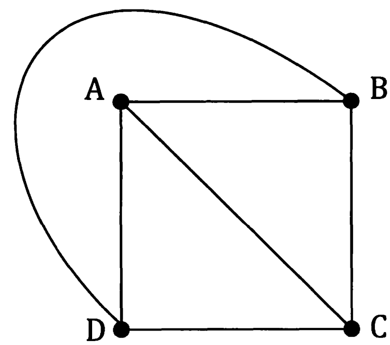
(c)



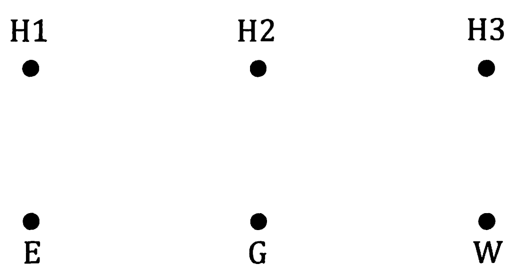
14. (a) Explain why each of the 3 graphs shown below are "complete" graphs.



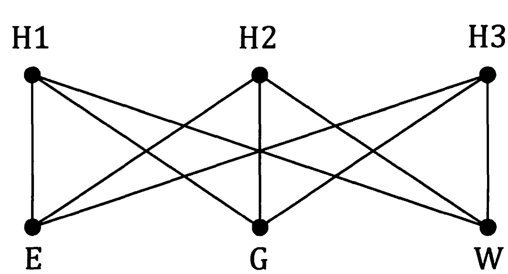
(b) For the first of the three complete graphs shown above it was explained some pages earlier that the graph could be redrawn with its edges only intersecting at vertices (i.e. without any edges "crossing over") as shown on the right.
Can this be done for the other two complete graphs shown in part (a)?



15. Suppose we have three houses, H1, H2 and H3 that the electricity company, the gas company and the water company want to connect up to their local connection points E, G and W.



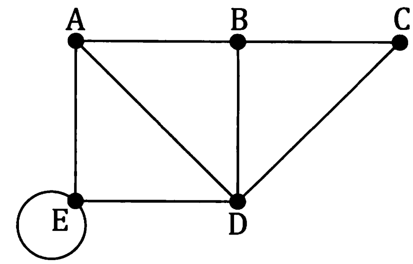
Each company wants to give each house its own dedicated direct connection.
One way this can be done is as shown below.
However, the companies would prefer it if none of the edges crossed. Can this be done? If not can you draw a network with as few "cross overs" as possible.



ISBN 9780170350464.

The degree or order of a vertex.

In an undirected network we say that the **degree** or **order** of a vertex is the number of edges meeting at that vertex. Thus for the network shown on the right,



vertex A is of degree (or order) 3,
(sometimes written deg A = 3)
vertex B is of degree (or order) 3,
vertex C is of degree (or order) 2,
vertex D is of degree (or order) 4,
and vertex E is of degree (or order) 4. (Each end of the loop counts, hence the order of 4.)

We say that A and B are **odd vertices**, because for each of these vertices the order of the vertex is an odd number.

We say that C, D and E are **even vertices**, because for each of these vertices the order of the vertex is an even number.

You may well have considered such things when deciding if a network was traversable earlier in this chapter because:

➔ **For a network to be traversable it must be connected and have either zero odd vertices or exactly two odd vertices.** 🡐

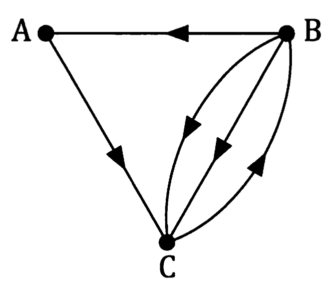
➔ If a connected network has zero odd vertices a traversable route can start at any vertex and will finish at that same vertex. 🡐

➔ If a connected network has exactly two odd vertices a traversable route must start at one of these and will then finish at the other. 🡐

Note: For *directed* graphs we have some edges coming into the vertex and other edges leaving the vertex. Thus we cannot simply talk of the order or degree of the vertices in a digraph. Instead we refer to an *in-degree* and an *out-degree*.
The "in-degree" of a vertex is the number of edges "coming in" to the vertex.
The "out-degree" of a vertex is the number of edges "going out" from the vertex.

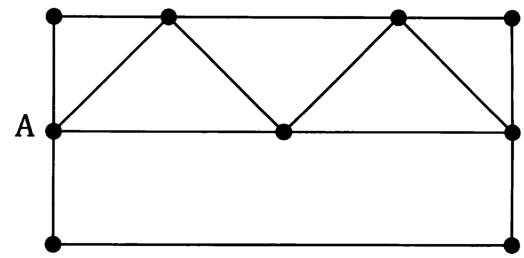
Thus for the digraph shown below,

vertex A has an in-degree of 1, and an out-degree of 1,
vertex B has an in-degree of 1, and an out-degree of 3,
vertex C has an in-degree of 3, and an out-degree of 1.



Two particularly important network situations.

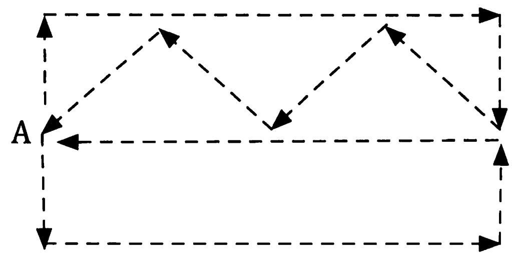
A system of streets is as shown on the right. Suppose you are asked to organise the route that a street light inspection vehicle should take so that it starts and finishes at the vertex marked A and passes down each street once.



This is a traversability problem. It's like the Konigsberg bridge problem – travel along every edge once and once only, revisiting vertices if necessary, and ending back at the starting point.

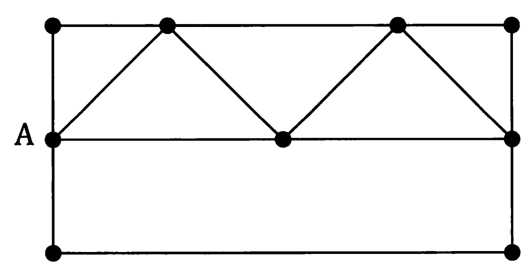
The vertices in this graph are all even vertices so we should be able to find a suitable route.

A possible solution could be:



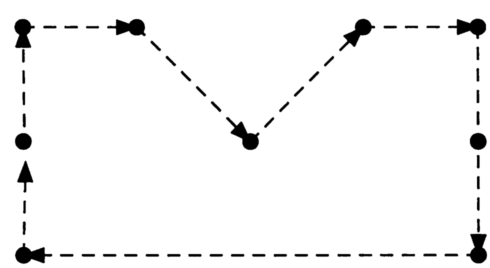
Contrast this with the situation in which the vertices represent the nine locations in a city that someone planning a walking tour wants to visit all of, starting and finishing at A.

Now we do not want to travel down all of the streets if we don't need to, we just want to visit every vertex.



This is the sort of situation a travelling salesman might encounter with A representing his office and the other vertices representing locations he needs to visit. He wishes to visit every location and end up back at A, without visiting any location twice (except for his office which he will start from and end at). Now it is the vertices that must all be visited. If it can be done without travelling along particular edges that's fine.

Now a possible solution could be:



In the first situation, involving the street light inspection vehicle, we were looking for a closed trail, i.e. one that finished at the same point it started from, that travelled every edge once and only once, vertices being visited more than once if necessary.

A **trail** in a connected graph that travels every edge once and only once, repeated vertices permitted, is an **Eulerian trail.**

☞ If the trail is closed, i.e. starts and finishes at the same vertex, it is said to be an **Eulerian circuit** and the graph is said to be **Eulerian.** Hence a graph is Eulerian if it is traversable, starting and finishing at the same point.
To be Eulerian a graph must have no odd vertices.

☞ If the trail is open, i.e. starts and finishes at different points, the graph is said to be **semi-Eulerian.** An open Eulerian trail is sometimes referred to as a **semi-Eulerian trail.** Hence a graph is semi-Eulerian if it is traversable, starting and finishing at different points. To be semi-Eulerian a graph must have exactly two odd vertices.

In the second situation, involving the walking tour, we were looking for a cycle that visited every vertex once and once only (except for the starting vertex that would be visited again upon completion). We did not have to travel every edge but did not want to travel any edge more than once.

A **path** in a graph that visits every vertex in the graph once only, with the possible exception of visiting the first vertex again at the end, thus making the path a cycle, is called a **Hamiltonian path.**

☞ If the Hamiltonian path is closed, i.e. begins and ends with the same vertex, the graph is said to be **Hamiltonian** and the closed path, or cycle, is said to be a **Hamiltonian cycle.**

☞ A connected graph that contains an open Hamiltonian path, but not a Hamiltonian cycle, is said to be **semi-Hamiltonian.**

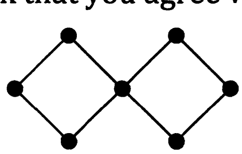
Want to travel every edge exactly once, and don't mind repeating vertices?
Think Euler.

Want to visit every vertex exactly once (with the exception of maybe revisiting the starting vertex at the end), don't mind if some edges are missed, but don't want to travel any edges more than once?
Think Hamilton.

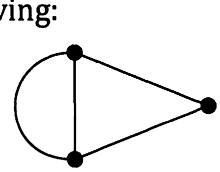
Leonhard Euler.	Swiss scientist and mathematician.	1707 - 1783.
William Hamilton.	Irish scientist and mathematician.	1805 - 1865.

Unfortunately, unlike the counting odd and even vertices that we can do when looking for Euler trails, there is no similar general test for the presence of Hamiltonian paths.

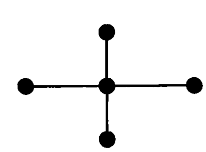
Check that you agree with the following:



Eulerian.
Semi-Hamiltonian.



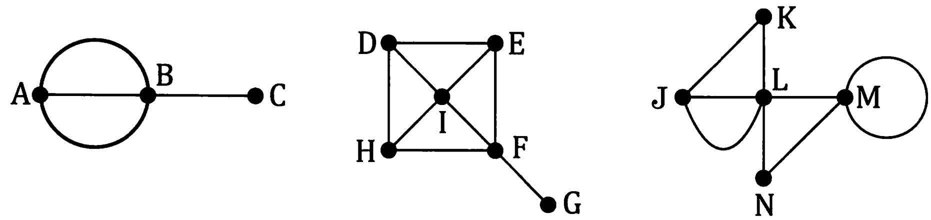
Semi-Eulerian.
Hamiltonian.



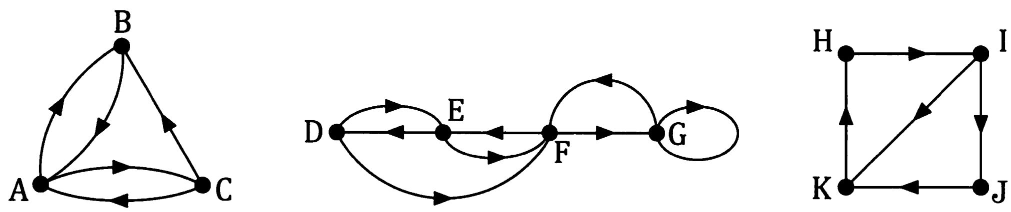
None of the classifications.

Exercise 5D

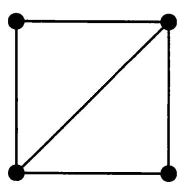
1. State the order of each of the vertices in the graphs shown below.

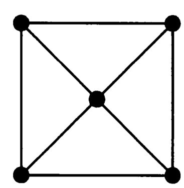


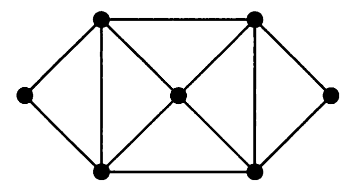
2. State the in-degree and the out-degree of each of the vertices shown in the following digraphs.

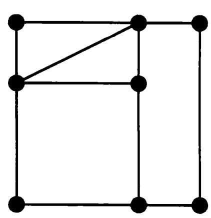


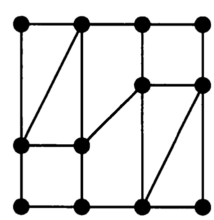
3. Classify each of the following graphs as Eulerian, semi-Eulerian or neither of these.

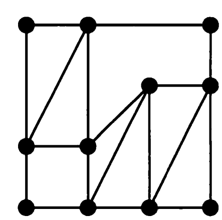
(a) 

(b) 

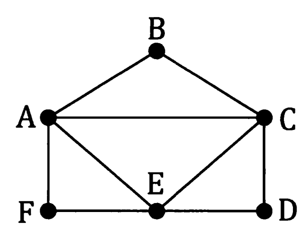
(c) 

(d) 

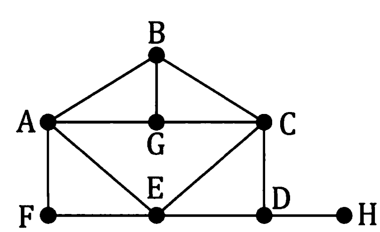
(e) 

(f) 

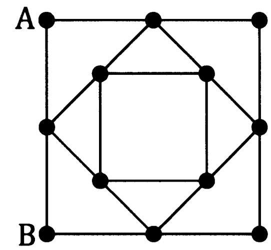
4. (a) Find an Eulerian circuit for the graph shown on the right, giving your answer as a sequence of vertices.
   (b) Find a Hamiltonian cycle for the graph shown on the right, giving your answer as a sequence of vertices.



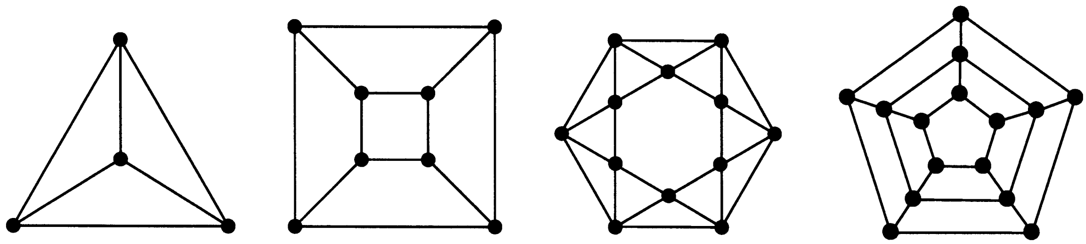
5. (a) Is the graph shown on the right Eulerian, semi-Eulerian or neither of these?
   (b) Is the graph shown on the right Hamiltonian, semi-Hamiltonian or neither of these?



6. (a) Sketch an Eulerian circuit for the graph shown on the right, starting and finishing at vertex A.
   (b) Sketch an open Hamiltonian path for the graph shown on the right, starting at A and finishing at B.

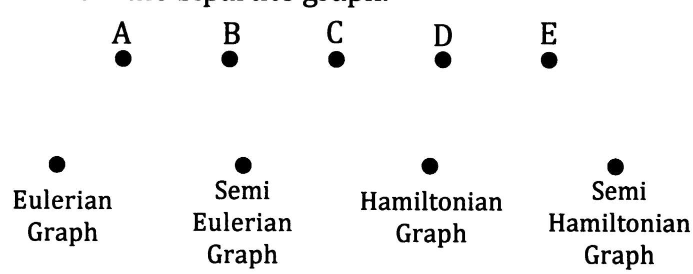


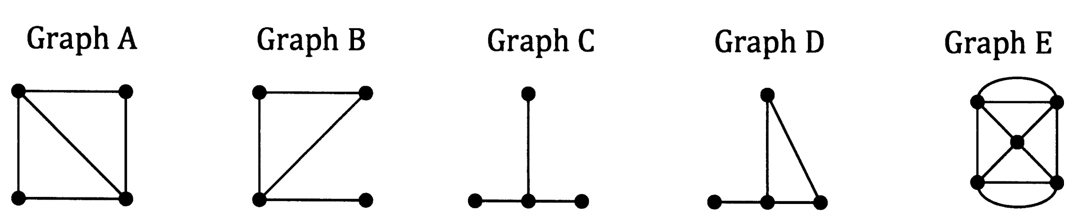
7. Find a Hamiltonian cycle for each of the following Hamiltonian graphs.



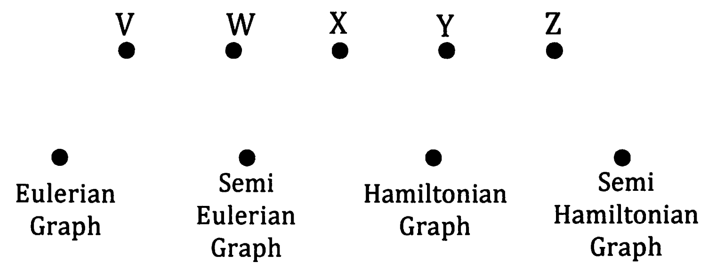
Note: In some texts the definition of a semi-Hamiltonian graph leaves some doubt as to whether a graph can be both Hamiltonian and also semi-Hamiltonian. However, if we use the definition of a semi-Hamiltonian graph given on an earlier page in this book a graph cannot be both Hamiltonian and semi-Hamiltonian. The next two questions assume this latter situation to be the case.

8. Construct a bipartite graph like that shown below, where A, B, C, D and E refer to the graphs shown below the bipartite graph.

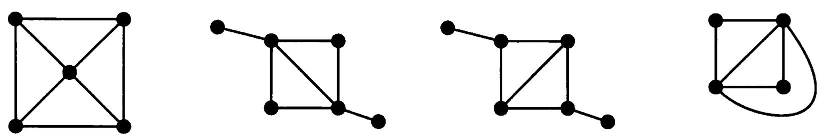




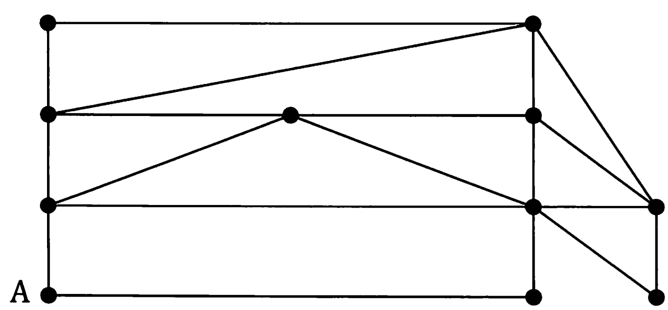
9. Construct a bipartite graph like that shown below, where V, W, X, Y and Z refer to the graphs shown below the bipartite graph.



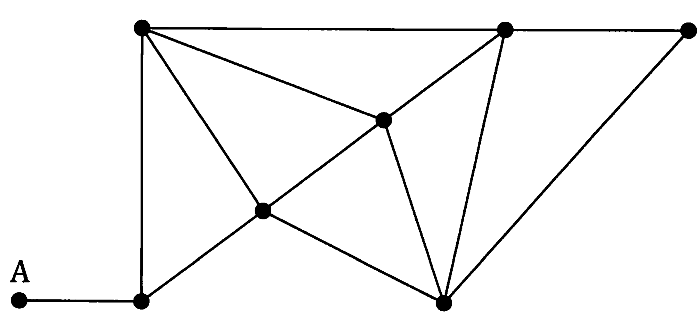
Graph V | Graph W | Graph X | Graph Y | Graph Z



10. Design a route around the road network shown below for a road inspection unit that needs to start at A, finish at A and pass down every road once and, if possible, only once.

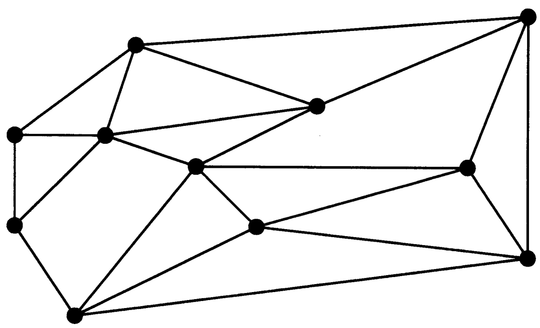


11. A contractor is given the job of re-marking the central road line on the system of roads shown below.



The contractor wants to design a route that starts at A, finishes at A and, if possible, passes along each and every road in the system once and once only. What route would you advise the contractor to take?

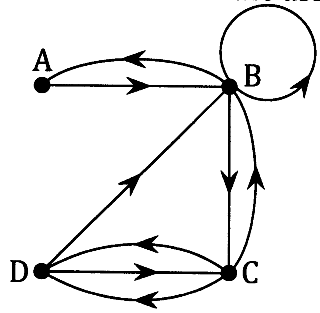
12. The organisers of a tourist brochure wish to show a circuit around the city that can be started at any of the eleven historically significant points in the city, takes in every one of the points and returns the walker to their starting point, having visited each of the points once. The graph below shows these eleven points as vertices and the pathways connecting them are as shown.



Design a suitable circuit.

Adjacency Matrices.

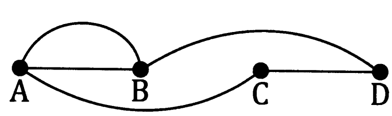
As the *Preliminary Work* section reminded us, the direct links in road networks and social interaction networks can be shown using a matrix form of presentation. For the digraph shown below left the associated matrix is as shown below right.



$$\text{From}\ \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \text{To} \\ \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

The entry in the second row and third column of the matrix, for example, shows the number of directed edges there are from vertex B to vertex C (in this case, 1).

Undirected graphs can similarly be presented in matrix form:



$$\text{From}\ \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \text{To} \\ \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

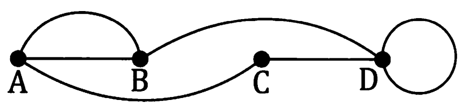
The two matrices shown above are examples of **adjacency matrices**. They inform us about the adjacency of pairs of vertices.

Notice that in the second case, with no loops present in the graph, the leading diagonal of the matrix shows only zeros, and with the graph being undirected the entries in the matrix are symmetrical across the leading diagonal

For a graph with n vertices the adjacency matrix will be an $n \times n$ matrix.

In a directed graph
- the entry in row i, column j is the number of directed edges there are from vertex i to vertex j.

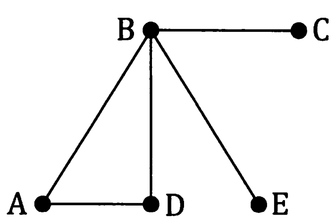
In an undirected graph
- the entry in row i, column j is the number of edges there are joining vertex i to vertex j,
- the adjacency matrix, A, will have $A_{ij} = A_{ji}$,
- a loop is counted as one edge. (See note below.)



$$
\begin{array}{ccc}
 & & \text{To} \\
 & & \begin{array}{cccc} A & B & C & D \end{array} \\
\text{From} & \begin{array}{c} A \\ B \\ C \\ D \end{array} & \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}
\end{array}
$$

Note: In some applications a loop in an undirected graph may contribute "2", as we saw with the order of a vertex. However there we considered the loop's "ends", rather like "how many ends has a piece of string?". Hence the contribution of "2". In the case of the adjacency matrix above we want the number of edges joining D to itself. There is 1 such edge. With edges undirectional it makes no sense to say the loop can be travelled "one way or the other". Hence the matrix shows 1, not 2. However, whilst we will use this convention in this unit, as in the unit glossary at the time of writing, it is not necessarily universal. In some fields of graph theory the convention for undirected graphs is to count a loop as 2 in the matrix.

Consider the graph and corresponding adjacency matrix shown below:



$$
\begin{array}{ccc}
 & & \text{To} \\
 & & \begin{array}{ccccc} A & B & C & D & E \end{array} \\
\text{From} & \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}
\end{array}
$$

There is 1 walk of length 2 starting at A and finishing at C. The walk is: A B C.
There are 2 walks of length 2 starting and finishing at A. They are: A B A and A D A.
There are zero walks of length 2 starting at B and finishing at E.

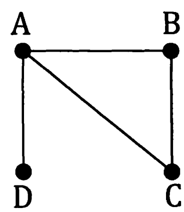
You may remember from *Mathematics Applications Unit One*, these "numbers of walks of length 2" can be obtained from the square of the above adjacency matrix.

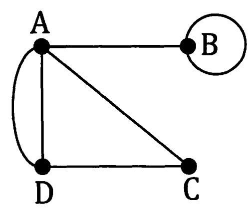
Use your calculator to confirm that the matrix shown on the right is indeed the square of the adjacency matrix shown above. Check that you can see where the numbers of walks of length 2 mentioned above can be found in the matrix.

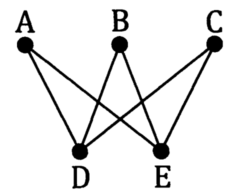
$$
\begin{array}{cc}
 & \begin{array}{ccccc} A & B & C & D & E \end{array} \\
\begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 2 & 1 & 1 & 1 & 1 \\ 1 & 4 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix}
\end{array}
$$

Exercise 5E

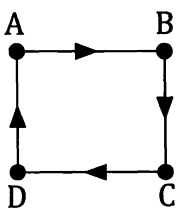
Write the adjacency matrix for each of the following undirected graphs.

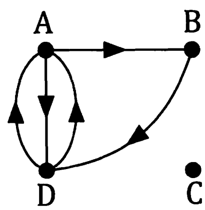
1.


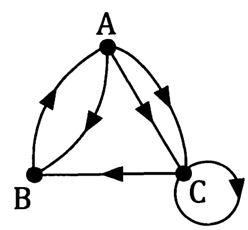
2.


3.


Write the adjacency matrix for each of the following digraphs.

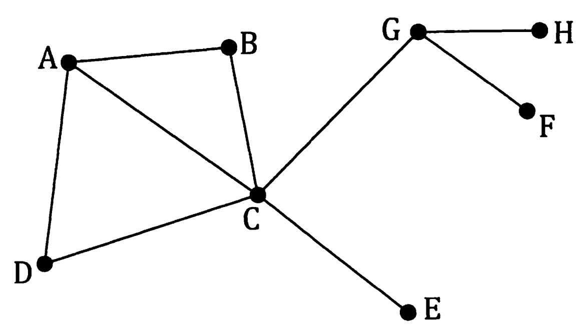
4.


5.


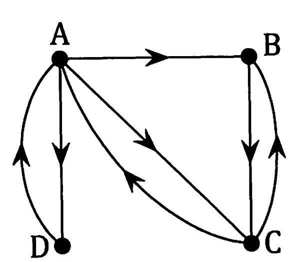
6.


7. We denote a complete graph involving n vertices as K_n. (Remember, a complete graph is a simple graph in which every vertex is connected to every other vertex by, in each case, a single edge.)
Write adjacency matrices for K_3 and K_4.

8. The diagram on the right shows the direct flight links an airline operates between eight cities A, B, C, D, E, F, G and H.
Write the adjacency matrix for this system and confirm that your matrix is symmetrical across the leading diagonal.



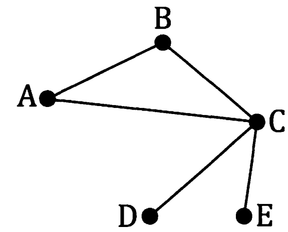
9. For the digraph shown on the right, and without using matrices, determine the number of walks of length 3 there are



(a) starting at A and finishing at C, listing each one as a sequence of vertices,
(b) starting at A and finishing at B, listing each one as a sequence of vertices,
(c) starting at B and finishing at A, listing each one as a sequence of vertices,
(d) starting at B and finishing at C, listing each one as a sequence of vertices.

Write down the adjacency matrix, M, for the graph and, using your calculator to determine M^3, check your totals for parts (a) to (d).

10. The network on the right shows the links between the cities A, B, C, D and E for which an airline operates direct flights. With a bit of thought you should be able to see from this graph that it is possible to fly with this airline between any two of these five cities in, at most, two flights.



   (a) Determine X, the adjacency matrix for this system.
   (b) Determine $X + X^2$ and explain how this confirms the earlier statement that it is possible to fly with this airline between any two of these five cities in, at most, two flights.

11. A national communications network has ten "hubs", A, B, C, D, E, F, G, H, I and J, through which various locations can be connected to other locations.
Not all possible pairs of hubs chosen from the ten have a direct connection between them.
In the adjacency matrix, X, shown below a "1" indicates a direct connection exists between two hubs, without the need for intermediate connections involving other hubs.

$$X = \begin{array}{cc} & \begin{matrix} A & B & C & D & E & F & G & H & I & J \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \\ I \\ J \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

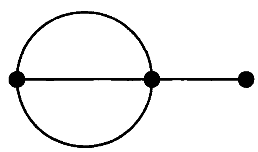
   (a) How can we tell from the matrix that there are no direct connections between hubs that are "one way only"?
   (b) Using a calculator that can handle matrix manipulation, or an online matrix calculator, and not by drawing the graph, show that a connection can be made between every possible pairing with in each case, at most, four inter-hub connections, and that some connections between hubs cannot be done in less than four.
   (c) Draw the graph and confirm that the situation described in part (b) is the case.

Miscellaneous Exercise Five.

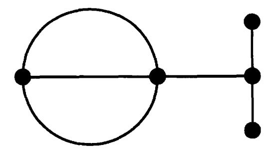
This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. For each of the networks shown below state whether the network is traversable or is not traversable.
   Which of the networks are Eulerian?
   Which are semi-Eulerian?

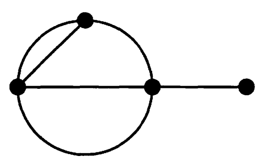
(a)



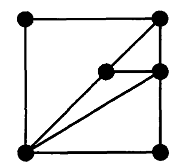
(b)



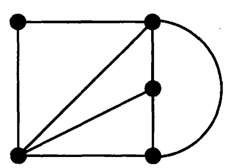
(c)



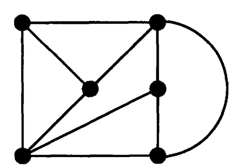
(d)



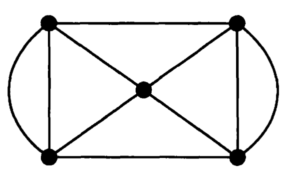
(e)



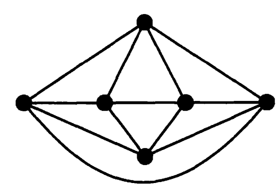
(f)



(g)



(h)



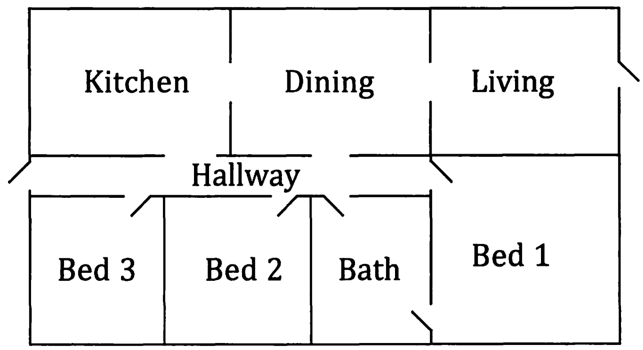
2. Is the sequence of account balances, in an account earning simple interest, an example of an arithmetic sequence, a geometric sequence or neither of these? Justify your answer.

3. Is the sequence of account balances, in an account earning compound interest, an example of an arithmetic sequence, a geometric sequence or neither of these? Justify your answer.

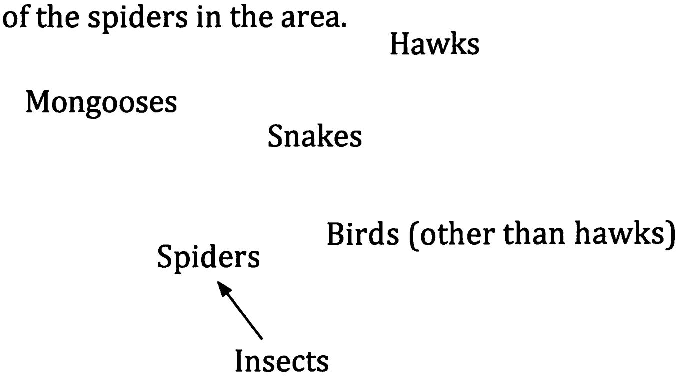
4. Find the first five terms of a sequence with a recursive formula:
$$T_n = 3T_{n-1} + 2, \quad T_1 = 5.$$

5. Find the first five terms of a sequence with a recursive formula:
$$T_n = 2T_{n-1} + 3, \quad T_1 = 5.$$

6. With each room, the hallway and the outside area as vertices and the direct connections between them as edges, represent the following house layout as a network.



7. A survey determined that the animals frequenting a particular forest area were mainly insects, spiders, birds (including some hawks), snakes and some mongooses.
The food web showing "who is being eaten by who" for the area has been started below with the one arrow showing that some of the insects in the area are being eaten by some of the spiders in the area.



Copy the food web and complete it using your judgement of "who is likely to be eaten by who".

8. In flat rate, or straight line, depreciation the value of an asset is depreciated by a fixed amount each year.
In reducing, or declining, balance depreciation the value of an asset is depreciated by a fixed percentage each year.
An asset that is initially worth \$64000 has a depreciated value of \$36000 two years later. What would be the value of this asset after one more year if the depreciation method used throughout is (a) straight line depreciation,
(b) declining balance depreciation.

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9. A commercial fish farming operation has approximately 10000 fish in one of its artificial lakes. It is thought that without any intervention this population will continue to increase at the rate of 1·5% per week.
However, intervention is planned that will see 200 fish harvested from the lake at the end of each week, the first harvest being one week from now.
Use a calculator that can cope with recursively defined sequences, or a computer spreadsheet to answer the following question:
How many weeks does it take for the number of fish in the lake to fall below 8000?

10. The table below shows P, the mean height for each of twelve pairs of parents, all who have at least one son, and S, the height of their eldest son when age 21.

P cm, mean parent height	168	179	175	179	161	183	170	172	174	186	175	163
S cm, eldest son height	170	186	172	185	162	191	181	175	179	199	176	171

(a) Determine the correlation coefficient r_{PS}, and the equation of the least squares regression line for predicting S from P.
(b) Use your equation from (a) to predict the height of the eldest son when aged 21 given that the mean height of his parents is 185 cm.
(c) Determine the equation of the least squares regression line for predicting P from S.
(d) Use your equation from (c) to predict the mean height of parents whose eldest son is 185 cm tall when 21.

11. Let us suppose that some mothers in two countries, A and B, whose eldest children were in the age range 12 to 16, were asked a number of questions about their eldest child. Further suppose that three of the questions asked were as follows:
Is your oldest child left handed?
Yes / No.
Would you say your child shows signs of being asthmatic?
E.g. wheezing, dry irritating persistent cough, shortness of breath, asthma medication.
Yes / No
Has you child ever received treatment at a hospital for any reason, other than at birth.
Yes / No
The results were as shown in the table below.

	Question about left handedness.	Question about asthma.	Question about hospital treatment.
Country A	93 Yes. 894 No.	177 Yes. 810 No.	149 Yes. 838 No.
Country B	234 Yes. 2179 No.	651 Yes. 1762 No.	191 Yes. 2222 No.

Investigate, analyse, write a report.